กระทรวงดิจิทัล เพื่อเศรษฐกิจและสังคม

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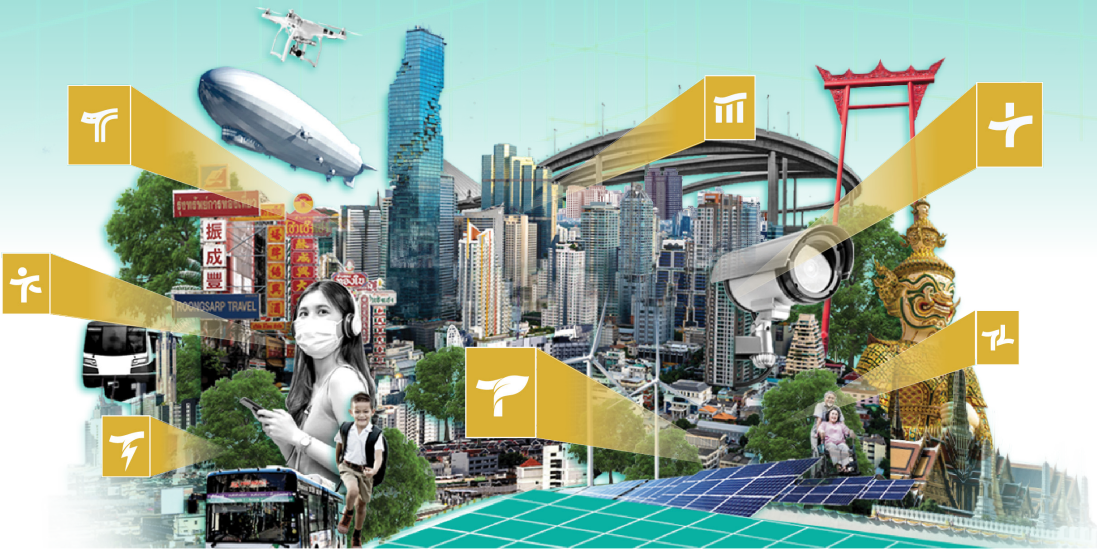
SMART CITY Thailand

ไทยแลนด์ SC20 W20

REAL Solutions for REAL People
in the REAL Smart Cities

16-22 OCTOBER 2020



EDITORIAL TEAM

Executive Editors

H.E.Mr. Chaiwut Thanakamanusorn
Minister of Digital Economy and Society

Ms.Ajarin Pattanapanchai
Permanent Secretary of the Ministry of Digital Economy and Society

Mr.Teeranun Srihong
Chairman of the Board of Commissioners
Digital Economy Promotion Agency

Dr.Nuttapon Nimmanphatcharin
President / CEO
Digital Economy Promotion Agency

Editors

Dr.Kasititorn Pooparadai
Senior Executive Vice President
Digital Economy Promotion Agency

Dr.Passakon Prathombutr
Senior Executive Vice President
Digital Economy Promotion Agency

Dr.Supakorn Siddhichai
Group Executive Vice President
Digital Economy Promotion Agency

Dr.Orachat Leingpeboon
Vice President for Smart City Promotion
Digital Economy Promotion Agency

Dr.Non Arkaraprasertkul
Senior Expert, Smart City Promotion
Digital Economy Promotion Agency

For enquiries, please contact

SMART CITY THAILAND OFFICE
(DIGITAL ECONOMY PROMOTION AGENCY)
Ladprao Hills building, 80 Ladprao 4 Alley, Chom Phon, Chatuchak, Bangkok 10900

Email:
scp@depa.or.th ,
info@smartcitythailand.or.th

Website:
www.smartcitythailand.or.th

Tel:
+66 2026 2333 ext. 5101-5111

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กระทรวงดิจิทัล เพื่อเศรษฐกิจและสังคม
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SMART CITY Thailand

Preface

H.E. Mr.Chaiwut Thanakamanusorn
Minister of Digital Economy and Society

The Ministry of Digital Economy and Society has played a leading role in promoting "Smart City" development in Thailand under the supervision of the National Steering Committee chaired by a deputy prime minister, and with the Digital Economy Promotion Agency (depa) as the Committee's Secretariat.

"Smart City Thailand" is a model for modern cities capitalizing on the use of technology and digital innovation to enhance the efficiency of city management and build a sustainable urban ecosystem. Aiming to improve the people's quality of life, the proactive and concrete action plan is grounded in several factors, such as the diverse needs of the local citizens, the readiness of resources, and the identity of each city. In doing so, the Ministry is considering robust methods, measures and mechanisms stemmed from productive synergy among sectors and long-term relations with trusted international partners.

Thanks to the prior commitments, this "Smart City Thailand" model is showing strong signs of progress, especially among cities that have recently been endorsed as future smart cities. For instance, the development of a tourism revitalization plan in Phuket Province, known as the "Phuket Sandbox," is a clear manifestation that integrates technology and digital innovation in different systems, including but not limited to the vaccine management system, public transportation, the application of 5G communication technology, the facial recognition system, as well as the internationally renowned safety system that safeguards international tourists in the city. Khon Kaen Province, also, is being promoted as a "Living Lab" with a clear objective to modernize health service technology and infrastructure. Another area worth highlighting is the Electricity Generating Authority of Mae Moh District in Lampang Province which has been designated and developed to become a clean energy hub for its community. Similarly, Wang Chan Valley, located in Rayong Province in the Eastern Economic Corridor (EEC), and the Samyan District in Bangkok are other key examples that are employing the use of technology and digital innovation, together with their vigorous roadmap, to become innovative smart cities.

To close, I am confident that the development of smart cities in Thailand will be a vital leap forward and become a fundamental mechanism helping to distribute prosperity to the locals, facilitates convenience, fosters public safety, and builds a strong and sustainable community going forward.

Preface

Mr.Teeranun Srihong
Chairman of the Board of Commissioners
Digital Economy Promotion Agency

The development of "Smart City" is one of the key themes of economic development and social transformation globally. In Thailand, Smart City refers to the mutual efforts and collaboration between the citizens—as the space owners, the government, and the private sector—to create a livable and modern city and enhance people's quality of life. As a crucial driver of this initiative, the Digital Economy Promotion Agency (depa) has implemented the "Smart City Plan" and aligned it with the 20-year National Strategy. Such a plan aims to build 100 smart cities in Thailand and drive three of them to be internationally accredited smart cities. At the same time, the Ministry of Digital Economy and Society has vigorously worked towards building efficient digital infrastructure to cover all of Thailand. Furthermore, as part of Thailand's 4.0 Digital Strategy, the development of smart cities is recognized as an essential apparatus of the Thai government to help reduce inequality and decentralize prosperity to other areas. As such, Thailand Smart City Week 2020 had been organized by taking into account the importance of connecting smart urban networks domestically and internationally. The purposes of the event were to navigate Thailand to becoming one of the future global smart cities, to bridge people and digital technology, thereby leading to sustainable economic development and social transformation. Last but not least, the event would manifest the persistent commitment and promptness of Thailand and its government in moving, with confidence and ample investment opportunities, towards the digital age.

Foreword

Dr.Nuttapon Nimmanphatcharin
President / CEO
Digital Economy Promotion Agency

"Thailand Smart City Week 2020" is the first hybrid event that encompasses the idea of smart city development. As a human-centric initiative, we have realized the potential of awareness-raising. Thus, we invite the participants to partake and directly experience digital technology via different online platforms available across regions. They would learn how to use digital technology, apply digital knowledge, build credibility, and bring investment opportunities into Thailand—all of which would manifest the zealous commitment of Thailand in embracing digitalization. Moreover, the smart city initiative will enhance modern technology and digital innovation to bolster the efficiency of city services and city management systems. Furthermore, the initiative will help save costs and maximize public resources, which will subsequently facilitate the cooperation of the public and private sectors. This can pave the way for people to live a good, happy, and sustainable life under the modern and livable smart city paradigm.

Testimony

H.E.Ambassador NASHIDA Kazuya
Ambassador Extraordinary and Plenipotentiary
of Japan to the Kingdom of Thailand

Due to the Covid-19 pandemic, citizen's lifestyles and economic activities have drastically changed all over the world.

In addition, nowadays ICT platform which is developed by applying new technologies such as AI and IoT is essential for daily life of the citizens, it is already recognized that the world is shifting to the life with "New Normal" such as working-from-home and online-shopping, and these trends have the potential to bring new light in solving various social issues in the future.

It is also required all urban functions such as transportation, commerce, business, medical care, energy, and government to be greatlytransformed to harmonize with digitalization by Digital Transformation (DX).

Thailand Smart City Week 2020 which was held last year under the leadership of depa achieved an outstanding success in capturing these trends and further accelerating Smart City Policy in Thailand.

Promoting Smart City Policy will make it possible to improve management system such as planning, development, operation, and maintenance by utilizing new technologies, to solve social issues that each city faces, and to implement sustainable city by creating new values.

Japan has been supporting implementation of Smart Cities in the ASEAN Region, especially in Bangkok, Chonburi, Phuket in the Kingdom of Thailand under the framework of ASEAN-Japan Smart Cities Network.

Japan would like to contribute to Thailand's economy by deepening Thailand-Japan relationship under Smart City Policy which will contribute to promote Thailand's National Policy such as Digitalization, Sustainable Development Goals (SDGs) and Bio-Circular-Green (BCG) economy.

Testimony

Ms. Alexandra McKenzie
Deputy Head of Mission of the British Embassy Bangkok

The British Embassy Bangkok is greatly honored to collaborate with the Digital Economy Promotion Agency (depa) on 2020 Smart City Week. Spearheaded by the initial launch of the UK-Thailand Smart City Handbook, our collaborative effort in promoting smart cities between the UK and Thailand exceled during the Smart City Week. The UK's partnership with depa continued with a series of Smart City Workshops, where the British Embassy Bangkok met with local authorities in Chiang Mai, Khon Kaen and Chonburi to explore potential areas for smart city collaborations. The partnership between depa and the Britis Embassy Bangkok led to the UK-Thailand Tech Export Academy virtual trade mission in May 2021, with over 200 business matching sessions between the British and Thai businesses.

The UK and Thailand are also working to deepen and strengthen its trading relationship across all sectors following the completion of a Joint Trade Review. In March 2021, the UK and Thailand signed a Memorandum of Understanding (MoU) which commits us to further strengthening trade and investment ties through a new joint trade dialogue. The Joint Economic and Trade Committee (JETCO) offers a significant new opportunity for the UK and Thailand to work together to improve bilateral trade, including by addressing market access barriers affecting businesses.

We value the professional and supportive partnership from depa's team and look forward to further collaboration with depa for many more years to come.

Testimony

Ms. Aileen Chew
Area Country Manager, Thailand and Myanmar, Mastercard

Reimagining cities and readying them for tomorrow's challenges is an enormous undertaking, especially in times of great uncertainty. Yet, that is exactly what Thailand Smart City Week 2020 set out to do. The tremendously successful event has given a voice to the thinkers who are engineering future-ready cities and a timely opportunity for the private and public sectors to drive meaningful change.

Through the event's live webinars that attracted audience from all around the world, Mastercard was able to share our vision and solutions for a Smart Thailand across many areas: from how blockchain can be used to support governance and efficiency in supply chains, to how technology can play a role in managing the movement of people in a post-pandemic world.

Beyond the event, Mastercard has been supporting depa with capability-building initiatives, such as educating and training for Smart City Officers, and ensuring the sustainability of turnkey projects across cities and municipalities through Public-Private Partnerships. Most importantly, depa is a proud member of City Possible, a partnership and co-creation initiative pioneered by Mastercard, that brings cities, companies and communities together to advance inclusive and sustainable urban development.

We look forward to continuing this momentum into 2022 and beyond.

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Thailand SC 20 W 20

REAL Solutions for REAL People in the REAL Smart Cities

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An Overview of Thailand Smart City Week 2020

The “Thailand Smart City Week 2020” is the first mega event that promotes smart city technologies and the first opportunity for Thai people to experience a REAL Digital Event in the field of Smart City. The event, which took place between October 16 to 22, 2020, aimed to drive smart city initiatives, make smart-city lifestyles more accessible, and encourage people to become vital drivers of smart cities. The Ministry of Digital Economy and Society, as well as the Digital Economy Promotion Agency (depa), play critical roles in steering Thailand toward Smart City development.

Thailand take advantage of modern technology and digital innovation, improve service efficiency and city management system, and improve people’s living quality under a modern and livable city paradigm. The success of this event is a milestone result of the mutual efforts between governmental organizations, the private sector, the education sector, and civic engagement. Furthermore, it is the first time that Thailand is focusing on developing a concrete idea of smart cities, which is stemmed from the REAL Solutions for REAL People in REAL Smart Cities concept.

There were four main pillars highlighted throughout the seven days of the event, in which the information is also available on scw.smartcitythailand.or.th

Smart City PLAYs

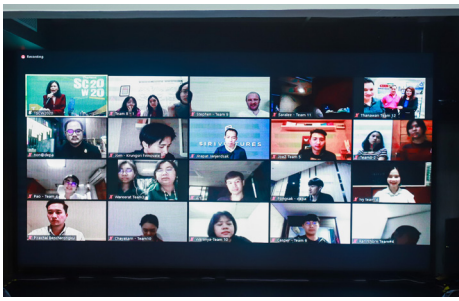
It is widely accepted that "playing" is the most productive and enjoyable way to learn. As a result, the Smart City PLAYs is a tool that demonstrates how we can apply technology and digital innovation in a real-world setting where participants can actually use it. Similar to the "Smart City Playground" concept, Smart City PLAYs is the result of collaboration between the public and private sectors, specifically the Samyan Smart City of Chulalongkorn University, Flower Labs of Silpakorn University, and Creator Space on the 6th Floor of True Digital Park—all of which are designed to increase public engagement and help tailor their smart city experiences.

Smart City LEARNs

Smart City LEARNs focuses on sharing activities, exchanging experiences, and expanding city-development knowledge through online seminars led by 24 experts from various fields. The training is intended to help people gain access to limitless learning resources and embark on a lifelong learning journey.

Smart City HACKs

A hackathon is a design sprint-style event that encourages technology and digital innovation with the goal of creating innovative solutions to urban city problems. The event is open to young innovators, who will participate in an intensive workshop with experts from various fields. The hackathon, which is described as a non-stop marathon-style competition, would require participants to come up with imaginative solutions to solve real-world urban problems. These solutions could be marketed for commercial purposes.



Smart City MEETs

Smart City MEETs is a virtual exhibition of smart city development hosted by more than 50 technology entrepreneurs. For the first time, the event gives Thais the opportunity to interact with smart city development technology from around the world. Furthermore, the virtual world-class exhibition hall would facilitate business negotiations among entrepreneurs, agencies, and business owners via chats and a real-time information processing system.

Highlights at

Thailand Smart City Week 2020

"Smart City Promotion Zones" Certificate Award Ceremony

During Thailand Smart City Week 2020, the Smart City Thailand Office was honored by Mr.Teeranun Srihong, Chairman of the Board of Commissioners of the Digital Economy Promotion Agency, who presented "Smart City Promotional Zones" certificates to representatives from 12 cities whose project proposals have been accepted by the Office. In addition to the 27 cities recognized during the previous "Digital Thailand Big Bang" event in 2019, these "second batch" of Smart City Promotional Zone include:

- **Samyan Smart City**
  by Property Management Office, Chulalongkorn University
- **Smart City in Rama 4**
  by Chulalongkorn University and ASEAN Smart City Network
- **Krabi Smart City**
  by Krabi Province

- **Saen Suk Smart City**
  by Saensuk Municipality
- **Saha Group Smart City**
  by Saha Pathana Inter-Holding (SPI) Public Company Limited
- **Nakhon Si Thammarat Smart City**
  by Nakhon Si Thammarat Province
- **Phang Nga Province to Smart City**
  by Phang Nga Province
- **Phitsanulok Smart City**
  by Phitsanulok Province
- **NU Smart City**
  by Naresuan University
- **Ban Chang Smart City**
  by Bann Chang Subdistrict Municipality
- **Wangchan Valley Smart City**
  by PTT Public Company Limited
- **Origin Smart City Rayong**
  by Origin EEC (Eastern Economic Corridor) Company Limited
- **Nongkhai Smart City**
  by Nong Khai Province

Smart City Promotional Zones are areas announced and designated by the Digital Economy Promotion Agency (depa) with specific objectives, namely:

1. To promote the use of digital technology and innovation to improve the economy, society, culture, and security in the agriculture, manufacturing, and service sectors;

2. To strengthen workforce development and digital citizenship in said areas; and,

3. To promote the use of digital technology and innovation to improve the economy, society, culture, and security in aforementioned areas.

In doing so, preparation activities include consulting on planning and implementation, technology selection, appropriate development guidelines for City Data Platform (commonly abbreviated as CDP), personnel development, and cooperation from domestic and international agencies.

The Launch of UK-Thailand Smart City Handbook

At Thailand Smart City Week 2020, the UK and the Thai Ministry of Digital Economy and Society jointly launched the "'UK-Thailand Smart City Handbook'." As the representative, Mr. Teeranun Srihong, Chairman of the Board of Commissioners of Digital Economy Promotion Agency, received the Hhandbook from H.E. Mr.Brian Davidson, the UK ambassador to Thailand. The Hhandbook's content covers an overview of smart city development in Thailand, along with examples of interesting UK smart city projects and expertise, including a series of collaborative activities between the UK and Thailand in promoting smart cities development. The ongoing projects consist of the following:

- The UK's Global Future Cities Programme with the Bangkok Metropolitan Administration to create an improved flood management system for Lad Phrao; to plan a Data Centre to support the collection and use of data for decision-making in Bangkok; and to create a development masterplan for the Khlong Bang Luang region and Bang Wa BTS Station. UK hosted smart city workshops in Chiang Mai, Chonburi and Khon Kaen provinces;
- Thailand is also chosen as the focal market under the UK's Tech Export Academy for Asia Pacific, which will see 30 British smart city companies undertake a trade mission to Thailand in 2021.

"I hope the launch of the 'UK-Thailand Smart City Handbook' and the on-ground expertise as part of the Digital Trade Network will provide the foundations for future cooperation, sharing of experience and expertise between the UK and Thailand in the areas of smart cities. We look forward to hosting British companies for the Tech Export Academy trade mission next year," said H.E. Mr. Brian Davidson, the UK ambassador to Thailand.

Those who are interested in the "UK-Thailand Smart City Handbook" can access it at:

UK – Thailand Smart City Handbook Launch
British Embassy Bangkok
depa

UK – Thailand Smart City Handbook
depa

UK – Thailand Smart City Handbook Launch
British Embassy Bangkok
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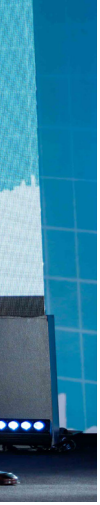

Joining Forces: Smart City Global Partnerships in Action

Cooperation with various partners is another critical factor that strengthens Thailand's smart city development and propels its global development path. Thailand has received extensive assistance from international partners such as MasterCard and other agencies from the United Kingdom, the United States, and Australia. As part of the ongoing partnership and commitment, each party has actively provided valuable perspectives and constructive discussions on the panel from:

- **Mr. Andrew Beirne**
  Economic & Prosperity Counsellor and UK Permanent Representative to United Nations Economic and Social Commission for Asia and the Pacific (UNESCAP)
- **Mr. Noah Geesaman**
  Deputy Economic Counselor, US Embassy Bangkok
- **Ms. Karen Surmon**
  Trade Commissioner & Counsellor (Commercial)
- **Ms. Aileen Chew**
  Country Manager of MasterCard

Moreover, Thailand had received immense support from governments, private companies, institutions, and the public sector, which is significant for devising innovative solutions and driving smart cities forward.

Thailand Smart City Week 2020's Success

Throughout the event, Thailand Smart City Week 2020 offered thought-provoking awareness of smart city development in Thailand to over 300,000 people, in turn generating more than 300 million baht from business negotiations. This has led to depa Thailand continuing with its excellent plan to promote the development of smart cities in 2021 through the "Smart City Ambassadors" initiative. This program has selected over 30 young individuals who have unique connections with their hometown and come from cities actively promoting smart city development. The main objective of such a selection is for talents from 30 cities across 26 provinces in Thailand to promote and enhance any essential skills required for smart and sustainable urban development. These activities will be initiated with representatives of city development agencies or the so-called "Captains."

The key activity is "Bootcamp," a one-year program that promotes livable smart cities through intensive masterclasses and fieldwork. The Bootcamp also aims to improve different dimensions of the cities in order to boost the quality of life of the people and bring forward innovative solutions to city problems. In doing so, the project will be supported by a network of digital startups and technology service providers. These providers will be involved either in the technological change for over 200 communities or no less than 2,000 business technology projects and the upskilling of the workforce, such as building new careers in artificial intelligence (A.I) and drone technology. Besides, the providers will help develop at least ten infrastructure projects in health services, tourism, transportation, public security, and the City Data Platform. As a result, investments valued at more than 500 million baht between the public and the private sectors will be contributed towards this project.

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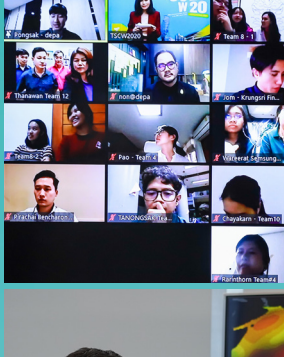

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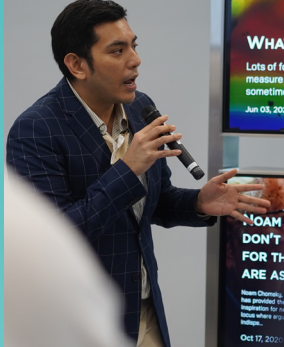


Smart City PLAYs

Why should you "experience" a smart city?

What difficulties or complications do you frequently face in your city these days? Whether it is a pollution problem, travel inconvenience, or waste and environmental problems, it is hard to deny that the more populated the city, the more **"problems"** will follow. Today, the word "smart" seems to offer an alternative for handling these collective city issues. **"Smart"** refers to using digital innovation and technology to help solve problems in a faster and more convenient way. With new management approaches, such technology has become a crucial and effective mechanism in handling and solving city problems. One realization that may contribute to the smart city development is that it is not only an obligation of government agencies or local authorities to carry out initiatives; instead, it is the mutual efforts of the private sector and the public that are of tremendous importance. Moreover, every sector should be resilient in its efforts to create a culture of openness to ideas, experimentation, and innovative digital technology. Besides, new management and investment models will eventually engender the sharp and concrete development of smart cities. Regardless of who you are, wealth, status, or career paths, every human is entitled to a good life with less social inequality and disparity. Achieving a good life may not

be an easy task and developing a smart city can be even more challenging. Yet, it is not an impossible mission.

Thailand Smart City Week 2020 transformed Samyan Mitrtown Hall into a space for activities like "Building a Tangible Smart City" and "Making the REAL Smart Cities," encouraging people to experience and broaden their perspectives of an actual smart city. The event was hosted by Wan Thanakrit Panichwit, a young, talented Host from the Prachachuen TV show, and Suchada Cotesin from depa Thailand. It included four simulating activities in 4 different zones:

• **Smart City HACKs:** Providing the opportunities for smart city innovators, Gen-Z and participants to share creative solution ideas through the Idea Wall;

• **Smart City LEARNs:** Offering online public training courses on smart cities; people have unlimited access to these lifelong learning courses anytime and anywhere to strengthen their knowledge in smart city development (registration is open via https://www.coursesquare.co/depa). This Smart City LEARNs is one of the mechanisms of the Chief Smart City Officer (CSCO) program, a key capacity building mandate of the ASEAN Smart Cities Network (ASCN);

• **Smart City MEETs:** Enhancing productive dialogues among city leaders, city developers, smart city offices, and technology service providers, and citizens;

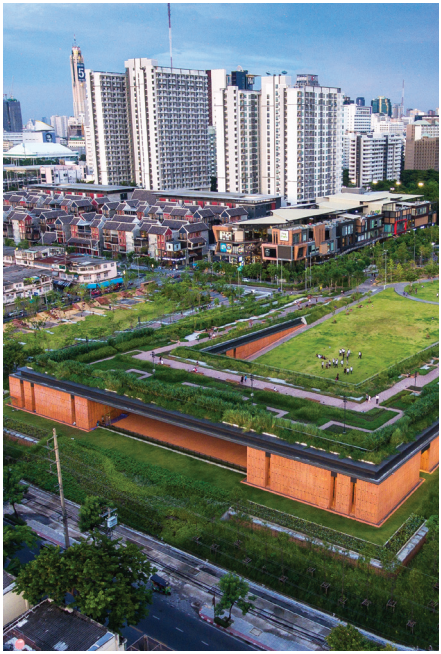
• **Smart City PLAYs:** Showcasing technological application to transform smart cities, improve the quality of life, and solve the city's problems. The platform is led by partner agencies of SAMYAN Smart City, including BANPU NEXT, MuvMi, HAUPCAR, Leaf e-Scooter, ViaBus, Impact Solar, TEAM Consulting, and UN-Habitat, to reinforce the idea of "tangible" or "REAL Smart City."

Opening to Technological Experience of the Smart City Development

To drive smart city development, access to all digital innovation technologies remains one of the essential requirements to all parties, particularly the central government agencies as well as local governments, the private sector, the education sector, and the public. Therefore, "Thailand Smart City Week 2020" is an important event that shares knowledge of technology and digital innovation and ensures the practical application in smart city innovation. As a result, the event is recognized as a tool to enhance the quality of life in various dimensions. Furthermore, participants would acquire hands-on experience in using technology to solve urban problems. In this context, Thailand's depa has cooperated with its partners both in public and private sectors to showcase the examples of 15 smart city areas—ten of which are located in Samyan.

- **Samyan Smart City**

By transforming the area of Samyan to become a smart city pilot, the space of at least 291 rai (116 acres) in the city center is required. This exciting transformation will develop Samyan into an innovation district creating shared values for the community and society and reflecting the unity of life and business. The ten locations surrounding the Samyan area are currently under the projects that follow the Smart Energy, Smart Environment, and Smart Mobility paradigms, as detailed below:



1. CU Centenary Park

Considered as the lungs of Bangkok, CU Centenary Park is an ample green space located at the center of the Suan Luang-Samyan area. The Park is covered by large trees, various plants, and several species of animals. Also found in the Chula Centenary Park, the famous pond acts as a recycling reservoir that redirects water purified by the wastewater management process at the CU Terrace & CU iHouse building to water the plants in the park. Furthermore, the air-quality monitoring devices will ensure that everyone can enjoy clean air while exercising as the air quality real-time monitoring and pollution detection systems give them both the peace of mind as well as the data-driven decision.

plication via CU Next, which allows you to effectively plan your trip using the EV Shuttle Bus's location-finding feature.

2. Mobility Hub

The Mobility Hub, located at CU Terrace and CU iHouse, provides a variety of transportation services to connect your journey before and after changing vehicles, also known as the "First Mile" and "Last Mile" journeys. Specifically, the hub offers electric vehicles (EV) as public transportation or EV Shuttle Bus, as well as EV "Tuk-Tuk" Sharing, EV Car Sharing, and E-Scooter. In addition, EV Car Rental services are available for travel between areas. Despite their disparate offerings, these modes of transportation share a common goal: to provide services using convenient and environmentally friendly electric vehicles. Users can access these services via the services' respective applications as well as a community ap-

3. Energy Hub

As the energy hub, Pathum Wan Station and Samyan Station are the primary substations of the Suan Luang-Samyan area, which are very important for electricity generation. Moreover, to supply energy to Chulalongkorn University, there is another Solar Energy project located at the top of the roof of Samyan Market (or on roof block number 28 and parking lot number 28). This project aims to generate a total of 13 megawatts of electricity production capacity by 2022.

4. Startup Village

A new central startup district is located on "Block 28," connecting creative lifestyles with a community emphasizing innovation and business connectivity. Such a district is an underlying factor in creating a startup ecosystem where several high-quality restaurants are being selected.

5. Public Service and Cashless Society

The promotion of Public Service and Cashless Society lies in the heart of Samyan Market—a fresh market that is highly required for its "diamond level" food safety standards and has acquired many awards. Moreover, as a public premise, the market remains one of the most necessary service facilities for the locals and non-locals to visit and enjoy their spending. Therefore, it has actively encouraged all the stalls to use cashless and contactless spending to improve convenience and the flexibility of the shopping experience. Similar public service facilities within the same area include police stations, fire stations, and health stations.

6. Samyan Creative District

Chula-Samyan, named one of the "40 Coolest Neighborhoods in the World" by Timeout.com in 2020, is an innovative area that encourages artists to express ideas freely by sharing community stories through art. Artists gather to help promote the neighborhood by displaying and distributing various artworks throughout the community in order to attract passersby, create new excitement, and boost economic activity in the area. Another highlight of this neighborhood is Samyan's art corners, which include Chula Art Town, Art4C, White Space, and the shophouse 1527xLabyrinth Cafe.

7. Chula Art Town

Chula Art Town signifies the urban arts surroundings or "Street Art." During the 36th anniversary of the Faculty of Fine and Applied Arts, Chulalongkorn University and the faculty—in collaboration with Chula Property Management Office (PMCU), Urai Commercial Company Limited (ATM SPRAY), and 50 most famous artists both domestically and internationally —came together to create street art for the community around Suan Luang-Samyan areas and Siam Square. It is the creation of art on the buildings and community walls around Chulalongkorn University that strongly mirrors the life in the community as an open-air museum that is accessible to all.

8. Siam Square Re-Landscaping

Siam Square's new look has become "wireless" thanks to the project that brought communication cables and power cables underground in the central commercial district. With the cooperative efforts between the Metropolitan Electricity Authority and telecommunication partners, the area has been transformed into a beautiful scenery with strengthened public safety for visitors.

9. Park @Siam

The small green space in the heart of Siam Square is a lung in the middle of the city covered with large trees and plants. In addition to reducing pollution, the space is open to the public with an easy-access location near a public transportation terminal.

10. Walkable City

A walkable city refers to a city designed around people and pedestrian activity to encourage the community's well-being. The project entailed building a 5-meter-wide pedestrian lane/walkway with an environmentally friendly atmos-phere within Chulalongkorn Centenary Road, i.e., Soi Chula 5. Besides encouraging people to walk more, the project also seeks to provide convenience and safe public spaces to create happiness and security among the community.

Samyan Smart City is an example of a "Smart Economy," or a city that uses digital technology to create value-added services in its economy, management, and efficient use of resources, such as smart agricultural and smart tourism cities. The essential purposes of the smart economy are to reduce the economic disparity of city residents, create economic opportunities for all groups of people and seamlessly integrate the supply and demand.

• **Flower Hub Space**

For the first time, Thailand has integrated art, technology, physical environments, and AR (Augmented Reality) together by utilizing software and other connecting devies. For instance, computers, smart cameras or images, videos, and sounds processed from computers, mobile phones, or wearable devices. Such integration will not only improve the process of purchasing flowers at Pak Khlong Talat, but it will also help connect the people and communities. Moreover, it can better consolidate the basic information/knowledge of flowers, leaves, and handicrafts inherited by the Thai community who have been associated with the Pak Khlong Talat Flower Market community. Last but not least, the integration will help create networking opportunities and innovation in the future. More details can be found at https://flowerhub.space/.

• **Creator Space (NEXT)**

Located on the 6th floor of True Digital Park Building, Punnawithi District, Creator Space is an innovative area that enables idea sharing, cultivates innovation, and helps create new startups. The space is jointly developed by King Mongkut's Institute of Technology Ladkrabang, Krung Thai Bank, and True Digital Park to generate idea exchanges and knowledge transfer on modern technology for young talents. Also, the Creator Space offers opportunities for the younger generations to experiment and truly work on building their startup businesses. They will learn how to develop solutions and solve problems through digital innovation, which has been an essential part of an urban and livable smart city ecosystem in Thailand.

At the same time, city leaders and city developers will visit the demonstration sites of digital technology and innovation—a collaboration between government agencies and the world-class technology leaders, which include:

- **Thailand 5G Ecosystem Innovation Center @depa**

Experience 5G technology at the Thailand 5G Ecosystem Innovation Center at depa Thailand, where innovations from laboratories and applications of 5G for businesses and services are unveiled, particularly 5G Medical Care, 5G Smart Agriculture, 5G Port, 5G Remote Education, and 5G Smart Security.

- **FutureTales Lab by MQDC**

FutureTales Lab is a learning space that focuses on future foresight and data analytics. Moreover, it creates a collaboration platform for futurists worldwide to engage in and reflect on the way of living, working, learning, leisure time spending, transportation, and sustainability. More information can be found at https://www.futuretaleslab.com/.

- **Huawei OpenLab Bangkok and Customer Solution Innovation & Integration Experience Center (CSIC)**

Located in the G Tower, Rama IX district, CSIC is Southeast Asia's most significant technology research and development center. By focusing on innovation development, collaborations, and new solutions between customers and business partners in Thailand and Southeast Asia, CSIC's focuses are public safety, Smart Grid, Finance, Education, Transportation, and Internet Service Providers. Besides, it also supports government policies, stimulates growth among Thai startups, and promotes Thailand's education system. If every province can adjust and fully develop into smart cities, Thailand would have higher leverage on information connectivity, well-developed public transportation systems, robust economy and technology, environmental benefits, and tourism growth. Consequently, this would encourage a higher level of social equality and make Thailand grow sustainably in the future.

PLAYS 12 จุด “เมืองอัจฉริยะ” ในกรุงเทพฯ

Chulalongkorn University

1. Block 28
2. Samyan Market
3. Centenary Park
4. CU Terrace
5. I-House
6. Suangluang Square
7. Dragon Town
8. Chula Soi 64
9. Siam Soi 6
10. Siam Soi 7

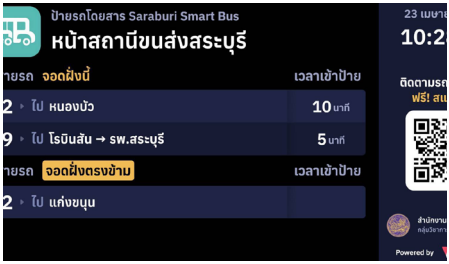
11 Flower Hub Space @Pak Khlong

12 Creator Space (NEXT)

SPOTLIGHT

“ViaBus” A Thai Startup that Responds to Lifestyles and Cracks City Logistics Problems

Public transport plays a vital role in the urban development agenda. Today, different forms of public transportation are offered in Thailand, including buses, minibusses, BTS Skytrain, MRT metros, express boats, and interprovincial vans. Undoubtedly, if everyone can equally and effectively access these well-integrated transportation systems, life in urban cities would be much easier, faster, and more convenient in planning and traveling. “ViaBus” is a platform that has brought about innovative ways of traveling in the country. Now an internationally recognized platform, ViaBus had been selected as one of the most-watched thirty innovators under the age of 30 by “Forbes 30 under 30 Asia – 2021.” As such, Mr.Intouch Maswongpakorn, CEO and co-founder of Viabus, also received the honorary award for Asians under 30.



Get to know ViaBus

ViaBus is Thailand's first transit tracking and navigation application that offers a rapid data processing or real-time system that connects all forms of public transportation networks in Bangkok, for example, buses, minibusses, BTS Sky trains, trains, express boats, commercial boats, vans, and interprovincial buses offered by the Transport Co., Ltd. The ViaBus services thus far have covered at least 70 provinces with more than 2 million users in Thailand.

However, the key objective of ViaBus is not only about expanding coverage areas or the number of users; it also aims to provide the precise locations of public transports, bus numbers, the nearest bus stations, and the fastest travel route. Moreover, provided that the application is available in both Thai and English, it serves as an efficient tour guide for both Thais and foreigners unfamiliar with the travel route. In addition, the application also displays real-time updates on relevant information, traffic conditions, canceling status, vehicle rest areas, or road blockages.

"More importantly, we also take into account the accessibility of public transport for persons with reduced mobility and/or with disabilities. With its unique features, ViaBus can specify vehicles with exceptional services, particularly for those in wheelchairs or visually impaired persons."

ViaBus's Journey

The beginning of ViaBus can be traced back to 2013 when Mr. Intouch had missed the bus during his freshman year at Chulalongkorn University. By missing the CU Shuttle Bus (commonly known as the "Chula Pop Bus"), Mr.Intouch and his friends were driven by the idea of tackling the waiting time paradox. Subsequently, he and his team jointly developed a Chula Pop Bus application and found that the number of application users skyrocketed, exceeding the whole population of Chulalongkorn University in less than a year. As a result, it has remained prevalent among shop owners, food sellers, and outsiders ever since.

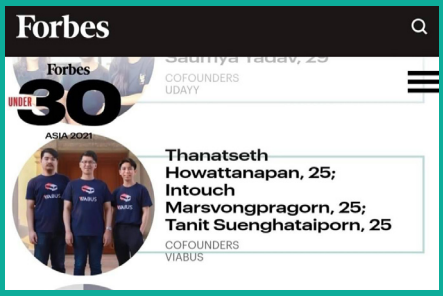
One justification for the booming of the Chula Pop Bus application is the fact that the application is user-friendly and has helped students and staff travel with convenience. Shortly after becoming a superstar of Chulalongkorn University, Mr. Intouch had foreseen its potential and decided to include new features to fulfill the extensive needs of the public. Two years later, the first version of the ViaBus application was launched with his other two friends—Thanatset Horwattanaphan and Thanit Suenghataiporn. Thanks to the immense financial support from different sectors, particularly the CU Innovation Hub's adept vision of new startups, ViaBus grew rapidly. ViaBus later formed new collaborations with depa Thailand, Ministry of Higher Education, Science, Research and Innovation, and the Bangkok Mass Transit Authority (BMTA).

The Next Step of ViaBus

In addition to the renowned ViaBus application, another successful initiative is called "Campus Program"—a system that tracks and navigates the routes particularly for university buses. The program is designed to specifically facilitate internal transportation systems, such as in large factories. Furthermore, it oversees the "Smart Bus Shelter" project, which offers real-time updates of the bus schedule and has now been installed in over 350 locations in Bangkok.

The underlying reason that contributed to the rapid growth of ViaBus over the years is that it can effectively address extensive problems and potentially fulfill the needs of users. To be specific, more than 99% of the ViaBus features have been developed based on real-life experiences/issues. As such, ViaBus represents a huge success that has brought about new impactful changes to society—all of which can be reflected from the awards we received over the years. These accolades shall remind us of how far we have come and remain a guiding light of where we want to go. That being said, ViaBus is merely the beginning of more impactful changes. Our next steps are to maximize the benefits for passengers and operators through planning, developing, and enhancing the transport ecosystem. Furthermore, we plan to connect every public transportation system and streamline it as one single system. We will increase the number and types of vehicles, such as local vans, and expand the application coverage areas to cover the whole country. This will encourage people to use public transport more, in turn alleviating traffic congestion and reducing energy consumption in Thailand.

ViaBus is a model of Smart Mobility, one of the seven areas of the smart city development plan. Smart mobility implies developing and improving traffic and smart transportation systems, convenience, travel, transportation safety, and environmental issues.



Smart City HACKs

4 Questions with Smart City HACKs Advisor, Dr. Supakorn Siddhichai

Why Do We Need a Smart City Hackathon?

When we try to solve a city problem, there are usually many stakeholders involved. So getting to the root of the problem is key. For example, when I was working on a project to reduce traffic accidents in a city. Our first job was to understand the nature of those accidents – these would usually include the cause, the people involved, the location, the time, and so on. Once these details have been established, we stepped back to look at the bigger picture to identify the heat map of traffic accident in that particular city. Perhaps we could prioritize them so that we utilize the available resources in the best possible way.

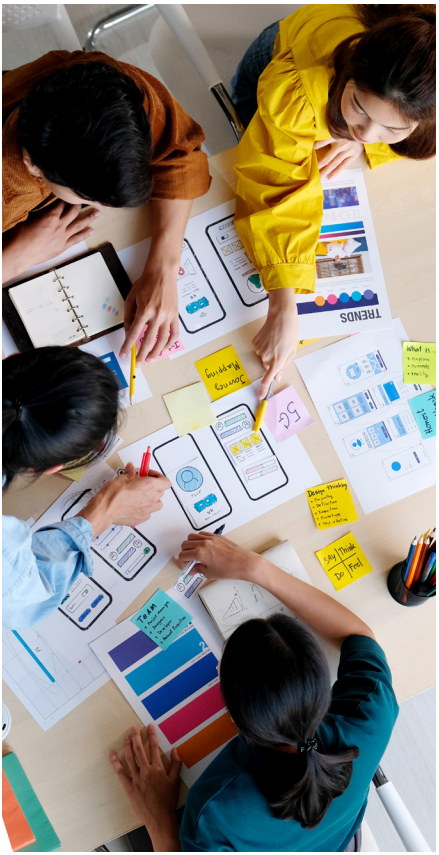
Then we explored what solutions were available to reduce or eliminate accidents at these top priority locations. In this particular example, we identified 10 locations and 5 causes – namely, red light running, bikers not wearing a helmet, illegal parking, not travelling on the proper traffic lane – or driving/riding in the opposite direction to the traffic flow, and last but not least, speeding. At each location, we installed different types of sensors to detect these traffic violations. And we implemented an automated traffic ticketing system where all the law enforcement officers need to do was to confirm the violations on the computer at the end of each day before printing out the tickets and sending them through the post. We all thought this would bring down the traffic accidents significantly. But what we didn't realize was that there was a law which states that each traffic ticket must be sent by a specially registered post and this cost more money. After 3 months of running the automated traffic ticketing system, the law enforcement offices ran out of money to pay for the postage.

What is the most important ingredient of a successful hackathon?

I wanted to emphasize that the user journey is extremely important to help us understand the nature of the problem. Also, a variety of ideas are crucial. The goal of a Hackathon is to come up with new ways to solve a problem. That's why we like to see participants with different backgrounds so we have better perspective to the problem we are solving.

Can you tell us a little bit of your background?

When I was working as a research scientist, my work was mostly involved with developing algorithms and software for image and video analytics. So, I got to work with medical doctors, law enforcement officers, and academics, on problems in their respective fields. Back in 2015, the Thai government launched smart city initiative and selected Phuket, our famous tourist destination, as the pilot site. I had an opportunity to lead a few projects there. Since then I've been involved with smart city development and promotion in other parts of Thailand with the Digital Economy Promotion Agency or depa.



Any words for the future hackathoners?

Fail fast and pick yourself up fast. So, keep testing your prototype and keep improving it. That's how I'd approach this Hackathon. As an engineer by training, I'm excited to see new ideas and solutions coming out of this Hackathon. I wish everyone a successful and enjoyable days to come.

"Hacking a Human-Centric Smart City"

Technology may not always be a panacea to smart city development, but it does play a significant role in simplifying tasks and demystifying conundrums for human beings. Despite such a fact, technology should be developed along with the "Design Thinking" paradigm that puts "people" at the center of the creative development process. People remain an indispensable factor to technology because technology can digitally sense, understand, and interpret human problems on a deeper level and unbiasedly than non-humans. Therefore, this is the main reason why depa Thailand, in collaboration with Techsauce, had jointly held the "Thailand Smart City Week 2020 Hackathon." The event lasted four days or 96 hours between October 15-18, 2020. As a challenging yet resilient virtual hackathon, it has been a good "test run" for all future activities that may need to fully embrace digital transformation and new everyday lifestyles owing to the ongoing situation the world is facing.

What is a "Hackathon"?

The term "hackathon" is derived from the words "hack," which means "to gain access to something new, agile, and innovative," and "marathon," which refers to a long-distance race. A hackathon, when combined, would imply an event in which people work continuously to develop the most innovative solutions, product concepts, or inventions. Participants in a hackathon, according to depa Thailand, would be required to brainstorm and develop innovations under a specific theme in a limited amount of time. The hackathon winner will be the team with the most innovative solutions/ answers.

"By hosting a hackathon, the organizers have realized the importance of time and the cost of waiting for new technology to transpire. Thus, a hackathon is simply a shortcut that can lead to creativity and rapid innovation development, which will be primarily required to address smart city problems."

The theme of this hackathon will bewas different from other forums since searching for solutions was is not the main priority, nor just simply putting citizen-centered agenda as its core approach. Instead, the participants must develop prototypes that couldan effectively tackle the city problems. With 70 teams participating in the event, the judges will only selected 12 teams from the first round to continue in the following stages along a 4-day competing journey.

Intense Competition, Full-House Experts

"Thailand Smart City Week 2020 Hackathon" comprised more than 15 professional staff, including speakers, experts, and consultants that engaged in knowledge and experiences exchange and thought-provoking workshops, namely:

- **Design Thinking**

  By Panunya Charoensawadpong

  Certified Design Thinking Expert from Stanford d. School Head of Business Development, AI and Robotics Venture

- **Business Model Blueprint**

  By Charle Charoenphan

  CEO and Co-founder of Hubba Thailand

- **Thailand Smart City: Understanding The Challenges and Opportunities**

  By Kungwan Laovirojjanakul

  Co-founder of Khon Kaen City Development (KKTT)

- **Talk Pitching Techniques**

  By Managing Director of SpeakPro Training

  Managing Director of SpeakPro Training

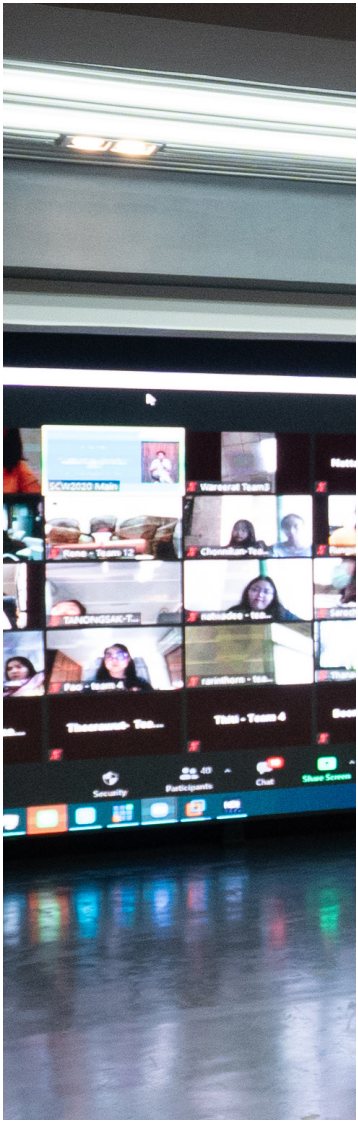
- **UX Design & Rapid Prototype**

  By UX/UI/Web Designer and Founder of Magnetolabs

  UX/UI/Web Designer and Founder of Magnetolabs

In addition, the event was joined by mentors —the experts in driving smart city development agenda, developers, forefront designers from leading companies, and potential investors. They offered helpful advice and recommendations to each of the 12 teams on a case-by-case basis so that the participants further built on such knowledge and polish their prototypes. Then, these prototypes would be used to target thecountry's problems and developed into a commercialized project. Expert consultants were:

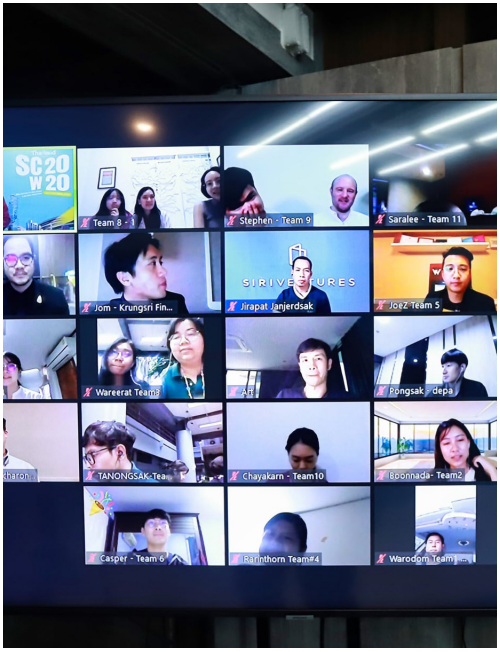
- Vilawan Layanun
  Service Designer of BridgeAsia
- Sitthiphorn Anthawonksa
  Head of CX of Kaidee
- Soranun Choochut
  Founder and CEO of etran
- Suradech Taweesaengsakulthai
  Co-founder of Khon Kaen City Development (KKTT)
- Amarit Jarasjindarat
  Lead Innovation Developer of SCG
- Warit Wanwithu
  Developer Relations of LINE Thailand
- Dr.Asama Kulvanitchaiyanunt
  CEO and Co-Founder of Coraline
- Kungwan Laovirojjanakul
  Co-founder of Khon Kaen City Development (KKTT)
- Kanpassorn Suriyasangpetch
  CEO and Founder of Ooca Telemedia
- Kanisara Kanjanawong
  Senior Business Incubation Manager of Beacon



The Last Day on Stage for Solution Mavericks

On the last day, each team present its prototypes to the panel of experts. The main criterion for determining the winner was to offer innovative solutions that had never existed before nor had ever been applied in any existing innovations. The winners also consider market competitiveness, business planning, and user-friendly design that prioritizes users' experience. Furthermore, the team operation should be resilient, convincing all the judges on the panel, consisting of:

- Dr. Non A
  Senior Expert in Smart City Promotion
  of Digital Economy Promotion Agency
- Mr. Pongsak Limpornpipat
  Former Senior Team Leader and Acting
  Director of the Digital Startup Institute at
  Digital Economy Promotion Agency
- Mr. Jirapat Janjerdsak
  Chief Technology Officer of Siri Ventures
- Mr. Kampanat Vimolnoht
  Head of Investment, Krungsri Finnovate
- Mr. Pirachai Bencharongkul
  Deputy managing director of United
  Technology Enterprise (UTE) and Investment
  Director of BCH VenturesSenior Business
  Incubation Manager of Beacon


Team 8 - 1
Stephen - Team 9
Saralee - Team 11
Jom - Krungsri Fin...
Jirapat Janjerdsak
JoeZ Team 5
Wareerat Team2
Pongsak - depa
TANONGSAK-Tea...
Chayakarn - Team10
Casper - Team 6
Warodom Team...

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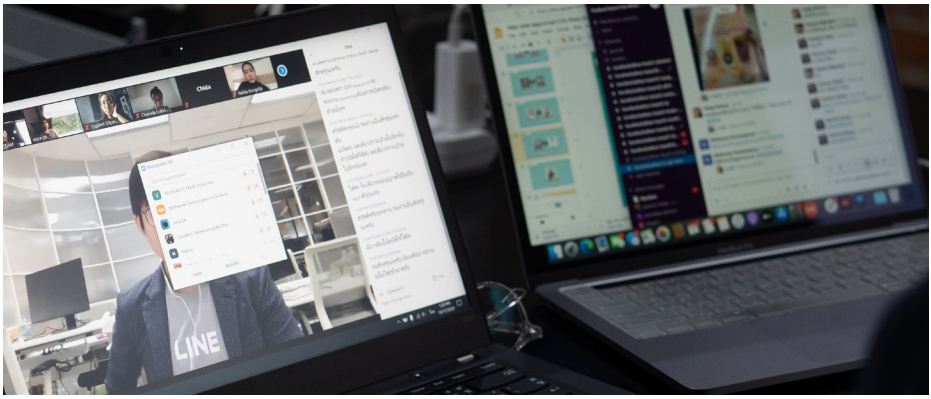
Jirapat Janjerdsak
Pongsak - depa

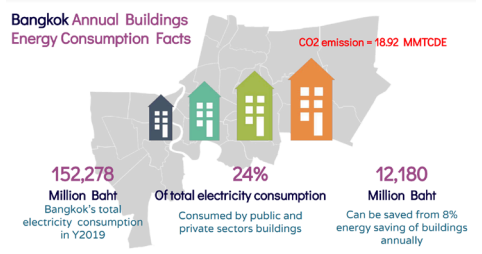

LINE

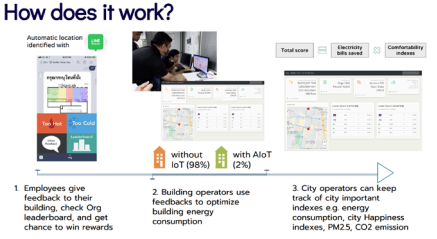
Finally, we identified the top three Thailand Smart City Week 2020 Hackathon teams, each of which received a combined 180,000-baht prize. The following are the winners:

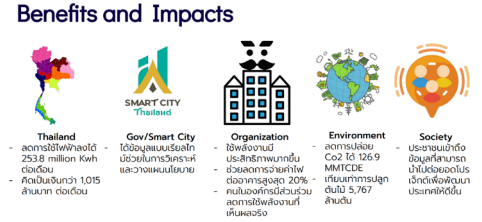
The Winner

AltoTech

Indoor Air Conditioning Energy Management System that engages all employees staff in the organization to interactively engage in the use of big data and artificial intelligence (A.I.). The team, in particular, used such technologies to monitor energy levels by allowing each building to compete to reduce energy consumption. During the competition, anyonefrom the public and private sectors can access the Smart City Center to check the energy -saving information of each building participates.







The First Runner-up

Workforce for Smart Labor Matching Platform

This team created a platform that provides training to people who are unemployed or whose jobs have changed. While the event's purpose is to facilitate customer seeking opportunities, it also serves as a mechanism to increase the business's efficiency level.

The Second Runner-up

The Tinder for Trash

This team built a platform called "Trashare" to connect environmentally-minded people and ensure that waste is being collected and managed in an appropriate and sustainable way.

In addition to the prize money, the winners also receive the opportunity to test their prototypes in real life by participating in smart city development projects. As a key supporter, depa Thailand will work with its smart city networks to apply the winning solutions to the problems in each city and help improve the quality of life of the residents.

SPOTLIGHT

Profits are expected by entrepreneurs in the business world. As a result, most of them focus on increasing production volumes, sales, and marketing in order to maximize profits from their investments. However, there is another way to increase profits while decreasing costs, particularly variable costs. Thanks to its volatility, would it be better for us to use technology and control energy consumption effectively? For example, would it be easier to calculate water usage and electricity bills by user-friendly, high quality, and affordable technology?

AltoTech, the Thailand Smart City Week 2020 Hackathon's winner, developed an innovative "Artificial intelligence of things" (or AIoT) Energy Management platform that helps manage energy consumption in hotels and office buildings. By combining Artificial Intelligence (A.I.) and the Internet of Things (IoT), hotels and office buildings could keep track of energy control, saving approximately 20-30 percent per month. The AI and IoT systems of AltoTech can be directly connected to air conditioning systems, chillers, electrical systems, CCTV of the building, including the solar cells generated power system. The primary purpose of such connections is to boost the efficient use of energy in each building and transform it into a Net-Zero Energy Building with Net-Zero Carbon Emissions.

Aside from that, AltoTech's platform can help building owners save money on operating expenses and equipment maintenance. In short, such innovation will transform ordinary buildings into smart buildings that can function autonomously by analyzing big data using Artificial Intelligence (A.I.) technology that is fully aware of the needs of buildings' owners.

Dr.Warodom Khamphanchai
CEO and Co-founder of AltoTech

"Smart Energy," one of the seven core areas of smart city development, promotes energy efficiency, reduces unnecessary energy consumption, and encourages the use of renewable energy.

Smart City LEARNs

Nowadays, the majority of people live in cities, where creativity and innovation are prevalent. Furthermore, urban areas are gathering places for various people, ideas, ideologies, and interests that can drive the economy on a regular basis. As a result, we can assert that innovation that has an impact on our way of life stems from the ecosystem of 'cities.' As a result, a city that promotes equality for its residents would benefit from cultivating a "homey" atmosphere, regardless of where the residents come from. Furthermore, an equal city would benefit the majority of people in a variety of ways, including improved quality of life, the economy, and society.

As a result, the concept of driving smart city development is analogous to having two sides of a coin. Residents will feel fulfilled and thrive in better living conditions if a city exudes the vibe of a modern and livable home. There will be fewer relocations. Technology can play an important role to solve the city's problems. As a result, the role of technology in this context entails the term "smart," which refers to technological intelligence. That said, technology that people does not want to adopt may be detrimental to the development of a socially sustainable urban community. To achieve sustainability, technology should eventually be aligned with a human-centered design and involve community participation from various sectors.

> **"Smart People are key to Smart Cities."**

People should learn, accept, adapt, and participate in designing and developing a city since no one knows what the city needs better than the local dwellers. This has led to the success of the Thailand Smart City Week 2020 Webinar, also known as "Smart City LEARNs," which encourages the participants to access new knowledge and ideas about urban development through online seminars with experts, thinkers, and city developers from Thailand and abroad. Before we move on to the Smart City LEARNs event, I'd like to prepare everyone for "The Fundamentals of Smart City," a free online course that shares basic knowledge in concepts and

methods of smart city development within and outside of Thailand, as well as the use of information in city management, including the analysis of the feasibility of popular projects. Those who complete the course will receive a certificate upon completion. Registration is open via https://www.coursesquare.co/depa.

Becoming a smart city does not imply having the most advanced and cutting-edge technology available. Rather, it could be a method that provides innovative solutions to the unique problems of each city. As a result, city leaders and developers must be open to learning new processes and embracing new knowledge in order to be aware of the stages of city development at all times.

Dr.Passakon Prathombutr
Senior Executive Vice President
Digital Economy Promotion Agency

One of the seven agendas driving smart city development in Thailand is **"Smart People."** It focuses on increasing knowledge and skills while also making the environment more conducive to lifelong learning. Furthermore, it seeks to reduce social and economic inequality while also encouraging creativity, innovation, and public participation.

The Right Way to Build a Smart City

Hearing perspectives of smart cities from 3 experts in urban development

Assoc. Prof. Dr.Jittisak Thammapornpilas

Assistant to the President Chulalongkorn University

If the city's residents only want a livable environment, a decent urban development should suffice. There is no need to "jump into becoming a smart city." On the other hand, if people believe they require a smart city, they must be aware of the investment required. Furthermore, "people" cannot be ignored because a city is built on the shifting paradigms of economy, environment, and sustainable growth. If we only produce based on ideas without considering "people," those ideas may resurface and harm the people themselves, rendering the city unsustainable in any way.

Chatpong Chuenrudeemol

Silpathorn Artist, Architecture of the Year 2020
Principal of CHAT Architects

Smart City Development should not be framed or sought by the best technology or a one-size-fits-all model to address urban problems. Instead, such a development should start from exploring the actual issues of the people in the area and examining the city's context. At times, being smart does not necessarily mean having clear boundaries around the city. Being smart can also be applied to one road or twenty-five families in a community. However, the most important thing is to learn, explore, and understand the people's real problems before searching for any solutions or applying them without knowing the potential contexts of the community. Thus, it is vital to understand a smart city from the eyes of those who live in the community.

Kwanchai Dumrongkwan

The Owner of the Bangkok Human Page

Cities are made up of people. A city is made up of many people who live together and participate in various activities. So, what constitutes a "Smart City?" First, you must investigate the issues that a specific group of people faces, because even the best and most innovative technology may not be designed to address specific issues or meet everyone's needs. However, what everyone requires is a tool that can help alleviate or resolve problems. For example, Bangkok City's problems are not linear; there is the complication of space, people, and sub-communities. As a result, listening to people's problems and tailoring the most appropriate solution(s) to their needs would be beneficial.

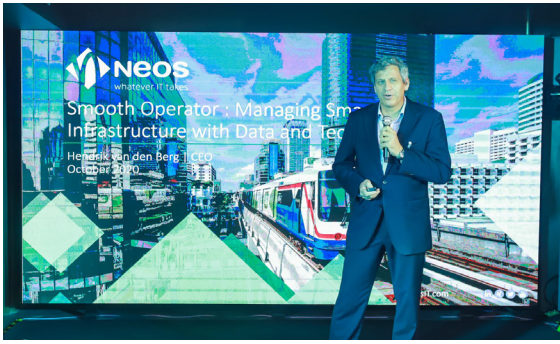
Smart Governance refers to a city that has an advanced governmental service system in place to meet the needs of its citizens. Residents have the right to equal access to government-provided sources of information based on transparency, active participation, and consistent improvement through the use of new technology and digital innovation as key stakeholders in the city.

4 Days, 24 Topics

Broadening new perspectives on smart city development

"Learn to be smart" with Smart City LEARNs and broaden perspectives on the development of smart cities from Thai and international experts. With more than 24 topics of knowledge sharing, the public can access the knowledge hub, and embark on a lifelong learning process journey via https://scw.smartcitythailand.or.th/.

DAY 1: International Outlook on Smart Cities Part I



URBAN MENUS Smart City Platform

Professor Laura P. Spinadel
Urban Menus (http://www.urbanmenus.com/)

"URBAN MENUS Smart City Platform" delves into the process of transforming Vienna, a city with arguably the best urban design and planning, into a smart city. The process employs computer programs, rather than the traditional method of design, to test city elements such as streets, buildings, and shops. Furthermore, the Platform investigates the feasibility of the prototypes and how a city can improve people's quality of life. To build a livable and economically influential smart city, it is necessary to have interdisciplinary knowledge of urban planning, socioeconomic development, and architectural design.

Smooth Operator: Managing Smart City Infrastructure with Data and Tech

Hendrik van den Berg
Managing Director, Neos IT Services

Neos IT Services is a well-known German company that creates a ground-breaking system that converts an organization's internal working arrangements to a digital-oriented system. The system educates users on digital transformation and advises them on potential steps that their organization can/should take. It does so by focusing on a reliable source of information, keeping the information up to date, and enabling feasible solutions/tools that help facilitate users' safe and efficient access to information.

Driving Contactless Payments in a Post-COVID World

Tobias Puehse
Vice President, Innovation, Mastercard

MasterCard, as a leader in cashless payment, is investing in its brand-new technology, which is currently in development. This innovation's underlying goal is to improve people's well-being through contactless information transfer. As part of one ecosystem of touchless experience, this in-development technology will allow users to witness the entire payment and service process, which can lead to a better economic system and a better environment—thanks to the convenience, transparency, and equal accessibility provided to people.

Digital Transformation and Innovation via International Partnership

Professor Alan Kin Tak LAU
Pro Vice-Chancellor of Research Performance and Development, Swinburne University of Technology

Professor Alan Kin Tak LAU, a researcher and executive at a leading university in Australia, indicates that the process of sustainable digital transformation consists of three major elements: ecosystem creation, device creation, and human development, based on extensive research, experimentation, and science-based hypotheses. As a result, educational systems must integrate research, development, and design in order to foster new opportunities for future A.I.-inspired smart cities, drive industrial market growth, and generate more environmentally friendly solutions.

How can cities recover “greener” and “smarter” after having been affected by the COVID-19?

Calum Handforth
Advisor, Digitalization and Smart Cities, United Nation Development Program (UNDP)

As the global development network of the United Nations, the United Nations Development Program (UNDP) views smart city development as a tool that encourages equal participation of everyone in every aspect of society. Through the application of “design thinking,” the cities are expected to grasp a better understanding of the underlying problems and subsequently address the real needs of the people. At the same time, the use of technology will help facilitate people’s access to the economy, health system, and education, which in turn, will lead to the reduction of social inequality and the increase in participation as the basis of economic growth and sustainable smart cities in the long run.

Smart City Ranking: Why Should Every City Care About it?

Calvin Chu
Managing Partner, Eden Strategy Institute

As one of the partner agencies of the Digital Economy Promotion Agency (depa Thailand), Eden Strategy Institute plays an important role in conducting research and consulting leading government and leading private sectors across the globe. The Institute is internationally known for its pioneering work on “Top 50 Smart City Government rankings” that explicitly focuses on the most important driver of smart city development or governments. During Smart City LEARNsS, Eden Strategy Institute had shed light on “why do we need to prioritize smart city rankings?” as a common strategy for each city would hinge upon its ways of development. Nevertheless, the aspects discussed during this lecture have increased our awareness of rankings, particularly

in the standard-setting process and the understanding of Smart City Ranking's mechanism. Undoubtedly, both factors can genuinely drive positive impacts in various dimensions of smart city development; The more international tourists we can attract, the better we can promote tourism and encourage confidence-building, leading to a more sustainable approach to developing smart cities.

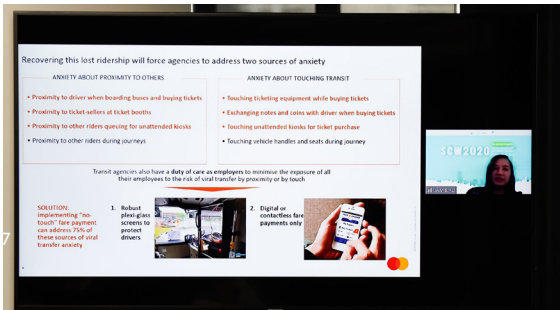
Private-Public Partnerships to Enable Smart Cities

Jeroen Kok
Director, Rebel Group
Shoubhik Ganguly
Principal, Rebel Group International, South Asia

Smart city investments take many forms, depending on the type of project. Municipalities, for example, frequently invest entirely in projects that are primarily concerned with the public's interests and are less concerned with profit-making. On the other hand, the private sector, typically represented by a city development company, would prioritize investing in profit-driven projects. Ultimately, "PPPs," or Public-Private Partnerships, are the third most well-known type of investment. PPP is frequently a collaboration between the private and public sectors, reinforcing each sector's comparative advantage. The public sector, in particular, would have greater clout on policy issues and legal proceedings. Despite its status as a central authority, the public sector frequently faces budget constraints, a lack of expertise in various fields, and a lack of incentives to develop a sustainable business. At the same time, despite having greater capabilities in terms of technology, digital literacy, innovation, and marketing, the private sector frequently encounters administrative bottlenecks from the government. As a result, the PPP is a perfect marriage of each sector's comparative advantage. PPP has been proven to deliver tremendous success around the world, according to two experts from the Rebel Group. For example, based on their experience, the design and development

process of a “smart pole” would necessitate. According to their experience, the design and development of a “smart pole” would necessitate advanced technological features such as wireless connectivity, online knowledge transfer, or the ability to detect PM2.5—all of which can be primarily supported by the private sector. Meanwhile, the government may help the project by enacting a taxation policy that allows for experimentation while also lowering production costs. As a result, as development, research, and technology become more competitive, the price will be forced to fall.

DAY 2: International Outlook on Smart Cities Part II



Making Tech Work for People Across the Globe

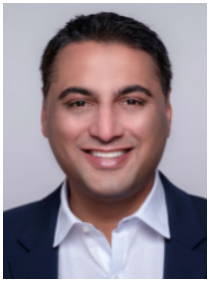
Miguel Gamiño Jr
Executive Vice President, Global Cities – Enterprise Partnerships, Mastercard

MasterCard is widely regarded as the world's second-largest credit card management firm. Today, however, MasterCard is striving for better transformation in order to avoid being disrupted by new technology brought about by new market players. Simultaneously, the company has positioned itself as a global tech company that sees opportunities in the challenges that cities face, such as health concerns and economic meltdown, which can primarily be addressed through multi-sectoral cooperation from all sectors around the world. As a result, MasterCard decided to collaborate with various organizations to implement projects such as donation platforms and contactless shopping technology.

Based on these motives, MasterCard has initiated the "City Possible" program that aims to create powerful technological tools to help improve the economy and the people's quality of life. In doing so, the project will assist the people in connecting with a global network of city leaders, industries, businesses, educational institutions, and government and private organizations across the globe. Furthermore, another project mandate is to encourage project members to exchange knowledge and innovation in particular seminars. Finally, several workshops will be held to constantly explore new challenges and examine the problems facing different cities in the hope of implementing the most sustainable solutions.

Miguel Gamiño Jr., Mastercard's Executive Vice President for Global Cities , has addressed issues and challenges global cities are experiencing. As such, it is the intention of the company that the City Possible project would serve as a third party that is highly committed to connecting global city members through information and resources, which, in

turn, will accelerate the pace of smart city development. Last but not least, Mr. Gamiño Jr emphasized that creating a smart city is possible through three major factors: a global network, urban co-creation, and a scale co-created solution or collaboration between tech providers.



Future of Work

Mona Mourshed
President & CEO of Generation

Poon King Wang
Director of the Lee Kuan Yew Centre for Innovative Cities National University of Singapore

Parag Mehta
Executive Director, Center for Inclusive Growth, Mastercard

People are becoming increasingly concerned about their jobs being taken away as technology advances. In reality, however, humans are constantly striving to improve and challenge themselves. As a result, claiming that new technology will replace human jobs is a fallacy. Work patterns must evolve in tandem with technological advancements. As a result, what remains critical is the development of skills and the acquisition of new knowledge in order to align them with new technology in the future.

Digital Transformation, Innovation, and Enterprise

Wachirachai Koonamwattana
Head of Living Solution Business, SCG

Office buildings in many cities usually consume more than 60% of energy usage. Thus, the acceleration of new and modern construction techniques can help reduce the problem of overconsumption with the support of the government. Notably, the government policy can monitor the cost of technology so that ordinary people can equally access it, potentially leading to sustainable city development. For instance, provided that buildings are primarily powered by electricity that releases a large amount of carbon dioxide into the atmosphere, SCG has presented a technological solution to air quality that is being released from air conditioners, which is to reutilize such air through the cleaning and monitoring systems.

Innovation for Life: An Inclusive Smart City on the Cloud

Kwei Foong Koh
ASEAN Smart Nation Business Lead, Worldwide Public Sector, Amazon Web Services (AWS)
Felix Tze Ghim Goh
Program Manager, AWS Educate, Amazon Web Services (AWS)

Cloud computing is yet another remarkable technology in city management. Because of the highly secure and accessible system, cloud computing allows users to choose whether to keep their data privately in a personal server, which prevents data leakage, or to open them up for public use, which can be accessed from anywhere. Furthermore, cloud computing can alter the size of data storage based on the nature of the data or the needs of the users. Furthermore, the system introduces five game-changing technologies that work well in the cloud and will be the future of a better world. These technologies are as follows:

Artificial Intelligence (AI): technology that enables computer systems to learn by themselves from machine learning and deep learning;

1) Augmented Reality (AR): technology that connects the actual world and the virtual world through various software systems and other connecting devices;

2) Internet of Thing (IoT): the connection of smart devices through the Internet;

3) Data Lake: mass storage with a better management system; and,

4) Voice Command: technology that connects cloud from Voice Control.

Taiwan Smart City Solution Showcase

Peggy Liao
Senior Sales Manager of FETC, Taiwan

The widely used e-Payment platform in Taiwan serves as a model for many cities. In particular, all types of public transportation in Taiwan can be paid for with a single card, simplifying all steps into a single streamlined system. However, implementing this type of system is difficult because it necessitates commitment, a desire for convenience, and a determination to provide equal access to customers and lower ticket prices. It also entails combining psychological and technological developments from both the private and public sectors in order to create this single platform.

How Blockchain Can Support Efficiency and Governance

Ashok Venkateswaran
Vice President, New Payment Flows,
Mastercard Labs, Mastercard

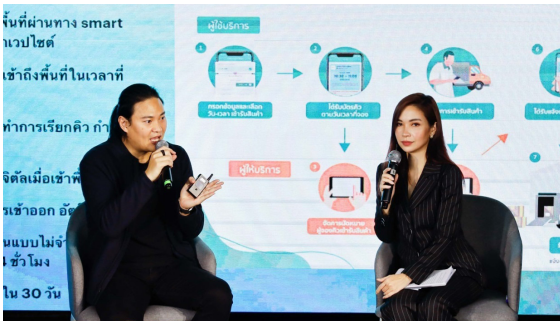
"Trust" is blockchain's keyword. As the number of internet transactions increases, blockchain technology adoption is critical for improving efficiency and building trust. Previously, we were used to an identity verification system that required a hard copy of an ID card; however, with blockchain, all personal information would be stored in a decentralized blockchain system, and only the owner of the data would be able to access and use such data. Consider how much easier life will be when trust is always guaranteed. The emergence of this technology will undoubtedly alter how we conduct transactions.

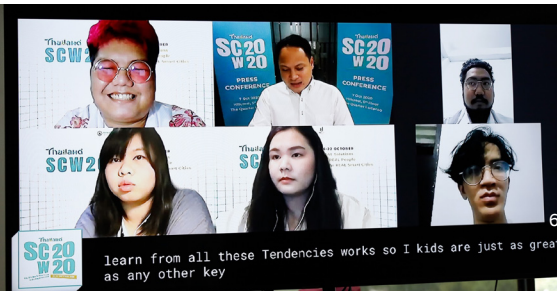
The Role of Technology in Realizing the new Normal for Public Transit

Jamie Fu
Vice President, Global Cities, Mastercard

Mastercard has introduced a technology that will employ blockchain technology in the public transport system. When implemented, the consumers will be able to access product information, trace the production line, and make a payment with the blockchain data. Moreover, the technology will improve public transport by installing a contactless payment system or e-tickets as one of the modern measures to help prevent the spread of COVID-19. However, provided that this blockchain technology is advanced and can be costly, not every city can afford it owing to concerns over investment, break-even points, local demand, and people's openness in adopting new technology.

DAY 3: The Local Context: Smart City for Everyone Part I





The Hands that Bring the Brightest Future: How Teachers Inspire the New Thinkers

Nopporn Insawang
Founder of Dynamic School Thailand

This session discussed how educational policies should be adjusted to remain relevant in the digital era. Today, the importance of institution names or grade point average has been replaced by the relevancy of digital literacy. Students should be equipped with digital skills so that they can use their talents to design and pursue career paths. Furthermore, students would be vulnerable to future trends or essential skill sets required in the labor market if they did not have strong digital literacy. As a result, the Dynamic School Thailand, as a modern classroom, will provide students with both theoretical and technical knowledge to prepare them for future transformations in the twenty-first century.

Growing Smart: The City for "Smart and Active" Aging

Dr. Warot Chotphithayasunon
Deputy Spokesman of the Ministry of Public Health

Assoc. Prof. Dr. Duangjai Lorthanavanich
Director of Ageing Business & Care Development Centre (ABCD), Thammasat Business School

Dr. Jaruwan Sripakdee
Department of Older Persons (DOP)

Thanakorn Phromyos
CEO & Co-founder of Young Happy Co., Ltd.

This session focused on exchanging views of smart city design for the elderly, particularly in terms of universal design, convenience, safety, and environmental concerns. Also, it encourages and supports the elderly to have good health and lifelong happiness. In part thanks to the advancement of technology and medical innovations, people can live longer with an average life expectancy of 77 years in Thailand, thereby making it essential for the elderly to have their retirement life planned, especially in the New Normal era where digital technology is a big part of our daily life. As such, raising awareness of digital media and proposing guidance on how to become 'smart' people with good health, recreation, integrity, cognition, and peacefulness remains a prominent agenda. Last but not least, support from government agencies, the private sector, and civil society is crucial to help the elderly adjust to an appropriate level of digital literacy and help cultivate post-retirement jobs, generate income, or facilitate access to health and safety services. Consequently, the elderly would be able to grow with confidence and stability in this fast-changing digital era.

Smart City: Smart อย่างไรให้เท่าเทียม

Manop Iam-saard
Manager of The Social Enterprise Development Center,
The Redemptorist Foundation for People with Disabilities

Suchart Owatwunasakul
Chairman of Disabilities Thailand (DTH)

Torphong Selanon
President of Thailand Association of the Blind (TAB)

Siriporn Praserdchat
Logistics Officer of Asia-Pacific Development Center on Disability (APCD)

Most people with disabilities have difficulty navigating public spaces, traveling, and communicating in their daily lives. As a result, when designing and building a livable smart city for all, smart city development should take these concerns into account. Four major elements must be reviewed and taken into account:

1) Smart Living – a place designed for people with disabilities to accommodate their activities and safety;

2) Smart Mobility - a public transportation system that supports people with special needs and helps them navigate their commute;

3) Smart Governance – a space that allows people with disabilities to participate in policymaking and decision-making process related to urban development planning; and,

4) Smart People – a concept that encourages the change in attitudes towards people with disabilities; in particular, it facilitates their access to technology, which in turn, can lead to employment opportunities in various ways.

Smart Gen-Z City: The Future is in Their Hands

Pet Journey (Sripatum University)
Expact (Thammasat University)
New Past Story (Sripatum University)

The activity introduces three presentations from 'Gen-Z youth' winners of Digital Ideas Contest under the theme of "Design Your Future, Gen-Z wants to tell." Smart Gen-Z City is a space designed for young talents to freely express their creative ideas, broaden perspectives on digital transformation, and build new inspirations. Moreover, it also helps those talents tailor their new career paths, opportunities, and smart cities for future generations. The first presentation belongs to the 'Pet Journey' initiative from Sirpatum University. The project introduces ways to connect applications to the tourism database to allow journey planning with your favorite pets. The second presentation is the 'Expact' project from Thammasat University that showcases how digital technology can help improve the teaching experience through an interactive learning system. Expact is expected to encourage and enable more children and youths to learn through the digital world. Last but not least is the presentation of the "New Past Story" project from Sirpatum University. The initiative applies digital technology in narrating the stories of Thai and other cultures. It aims to communicate with future/aspiring talents and encourage their learning experience in the field of culture and history of Thailand through online channels.

Startups x Cities: Pain Points to Solutions

Rungsun Promprasith
CEO of QueQ Thailand

The ideas of the QQ's application manager can be used to build a smart city in 3 areas. The first idea is related to the application design or software. Initially, the QQ was designed to manage queues at restaurants. However, in light of the pandemic, the application has since shifted its strategy, or PIVOT, to also operate different kinds of appointments, such as embassy appointments, hospital appointments for the treatment of patients, and visitors' appointments at government offices. This shift of strategy turned out to be highly beneficial in reducing the risks of contracting COVID-19.

The second area is related to solution finding or designing products and services that address pain points. This focuses on the "People First" principle, product testing, and prototype creation, which may take several stages to develop and become the most appropriate models.

The third topic is how to behave like a "pirate." This refers to how a startup use technology to bypass the red tape-filled process by which government agencies and the private sector work in concert to solve city problems.

Open House: A Call for Accessible Architecture and Urbanism for All

Chatpong Chuenrudeemol
Silpathorn Artist, Architecture of the year 2020 and Chief Executive Officer CHAT Architects

Architecture is an essential element of urbanization. The role of architecture in urbanization cannot be overstated. Most people think of architecture as the creation of new structures, such as oddly shaped buildings or the distorted use of space. In reality, architecture is a mindset or inspiration influenced by our surroundings that can aid in the creation of designs. Vernacular architecture, for example, is a style of architecture built with local materials, local wisdom, and local people's hands. It places a premium on craftsmanship and design to meet the needs of the residents. Furthermore, vernacular architecture fosters community participation, activity engagement, and human connection.

Therefore, becoming a smart city does not imply several buildings; instead, it focuses on every dimension in a city. Architecture is about connecting the community and patterns of life for the people. In the eyes of Mr. Chatphong, who led design research called "Bangkok Bastards" or "Uncompromising Architecture," he noted that the architecture of impermanent housing, such as those of construction workers, slum communities, or illegal markets, are stemmed spontaneously from "city people." Often regarded as the garbage dumps, the sad reality is that they do not have access to the urban economic cycle, and they must struggle for their survival, living on leftover, cheap materials like scrap metal, zinc, old wood, campaign signs, Fanta posters, or pictures of celebrities that are cut into wallpapers. However, from an architectural perspective, it is the art of recycling that serves the most meaningful purposes. Mr. Chatpong emphasized that as you delve deeper into this type of story, you will discover pure magic in these people's spatial arrangements. It is a wonderful realization that cities belong to everyone.

Social Entrepreneurship: Bridging Business, Society and Happiness

Peangploy Jitiyatham
Co-founder of Locall Thailand
Winyoo Srisupachok
We are Volunteer English Bangkok

The link between people, communities, macroeconomics, and national policies can help generate businesses within a community. "Locall" is an excellent example of a charitable business that provides delivery services to local food stalls. Despite several challenges in pricing, competitiveness, and quality, Locall, with the help of other agencies, has helped fulfill the growth of local food providers and sustain their businesses during the pandemic. Another example is "We Are Volunteer English Bangkok," which enters the community and assists locals in managing and balancing their school, leisure, and family lives. Based on these two examples, it is critical for any community-based project to thoroughly understand the needs of the locals and be prepared to address the city's pain points.

DAY 4: The Local Context: Smart City for Everyone Part II

Next-Gen Agro-Commerce: Taking Thai Agriculture to the Next Level with Digital Economy

Dr. Preesarn Rakwatin
Vice President for Digital Agriculture Development and Promotion of Digital Economy Promotion Agency

Krittha Bhokharsathit
President of E-Commerce Committee of MOAC

Thanain Toomhiran
E-commerce and Online Marketing Consultant

The COVID-19 global pandemic has profoundly caused labor shortages in the agricultural sector due to the repatriation of migrant workers. The farmers are forced to adopt the Internet of Things or IIoT technology an innovation that embeds electronic circuits, software, sensors, and network connections in digital devices and equipment. By doing so, the farmers can utilize the IoT to create automatic watering systems, automatic power on and off systems, and help distribute agricultural products via an accessible and affordable online platform, such as Facebook Live. Digital Economy Promotion Agency, as a critical stakeholder in smart city development, has also implemented several supporting measures, such as the depa Mini Transformation Voucher, which allows farmers to purchase digital products or services and assist them in applying modern agricultural techniques to their traditional farming methods.

Tackling Crisis like a Pro: How Medical Technology Take Us Through and Beyond COVID-19

Assoc. Prof. Dr.Cherdchai Nopmaneejumruslers
Vice Director of Siriraj Hospital

Dr.Chokchai Suwanakijboriharn
Advisor of Bangkok Dusit Medical Services (BDMS)

During COVID especially, medical technology has significantly helped Thai people provide public health services. For example, Siriraj Hospital has developed a drug dispensing system using automated robotic technology, a drive-thru blood collection service, and a telemedicine service for patients with mild symptoms via video call. For the latter, medicines would be distributed to the patients after the consultation session. In addition, the system also helps facilitate the patients who have been treated or diagnosed abroad. As we progress, the future of the medical and public health industry will be using artificial intelligence (AI) technology to do a preliminary assessment of symptoms and diagnose disease with fewer errors and a higher level of precision.

Beyond the Cash World: Digital Currency and the New Opportunity

Jirayut Srupsrisopa
Founder & Group Chief Executive Officer of Bitkub.com

In the future, money will no longer be paper. Instead, it will come in digital form. People will no longer have to carry cash, and they will be able to complete transactions via digital systems, as can be seen from Bitcoin's blockchain system that has transformed the world's financial system. Furthermore, banks in some countries have already stopped offering bank account services to their citizens, owing to the high costs of management, building a branch, and installing ATMs. With the emergence of digital currency, transactions can now be done via mobile applications, thereby allowing more people to easily access loans, personal accounts, or complete financial transactions anywhere and anytime.

Turning a Crisis into an Opportunity: Living in the Changing Post-Pandemic World

Dr.Udomthipok Praikaset
Vice President of Thai e-Commerce Association

Assoc. Prof. Dr.Peter Ractham
Associate Dean for Research and International Affairs, Thammasat Business School

The pandemic crisis is an accelerator of the rapid decline in the Thai economy. In particular, the tourism industry--which is the most significant contribution to the economy--has been severely hit by COVID-19. On the one hand, retail businesses, fresh markets, grocery stores have been deeply affected while on the other hand, the e-Commerce business is growing exponentially, thanks to the behavioral change of the consumers and the availability of online platforms. At the same time, we are witnessing the growth of the IT business as it highly fulfills the needs of work-from-home lifestyles. With these examples, entrepreneurs are encouraged to look for more opportunities in crises, adopt new strategies, and take advantage of each moment not only for survival, but also for a more thriving future.

SPOTLIGHT

"A city belongs to everyone, whether you are a child, a working-age person, an elderly, or a person with disabilities. Therefore, everyone must build their smart city because central to this development is the fact that the city must address the common problems and meet the needs of all people living in the city."

Turning Gen-Z into Digital Citizens

It is vital to start building and developing the digital workforce to support the country's development process. Yet, at the same time, it is even more significant to bring out the best potential of the so-called 'Gen-Z' talents, who are the hope of inclusive and equitable digital transformation and make the most of those capabilities through the creation of digital awareness and fundamental digital literacy.

depa Thailand, under the supervision of the Ministry of Digital Economy and Society, together with GU-Creative and partner agencies, had launched innovative products for Gen-Z people at the "Digital Your Life 2020: Digital World and the Lifestyles of the New Generation" event, moving forward with its commitment to cultivating digital citizens of the 21st century. The event aimed to promote digital literacy and propel the youth into becoming digital citizens who can use technology in a creative, beneficial way which, in turn, will improve the quality of life of many people in the city.

Another noteworthy activity is the "**Digital Idea Contest: Design Your Future, GEN-Z Want To Tell,"** an activity for youngsters to pitch ideas/solutions and share perspectives on country development in different fields, such as education, agriculture, health-beauty, tourism, and culture. Moreover, these young talents would be able to participate in Gen-Z MEETUP activities so that they can sharpen their digital skills and pick up great advice from digital experts. Eventually, there were three best ideas selected from 54 teams, consisting of the following:

The Winner: The Pet Journey Project from Sripatum University introduces the idea of using applications to connect with the tourism database, which can lead to the creation of a traveling journey with your favorite pets in the future.

The First Runner-Up: The Expact Project from Thammasat University presents how digital technology can help improve the teaching experience through an interactive learning system. Expact is expected to encourage and enable more children and youths to learn through the digital world.

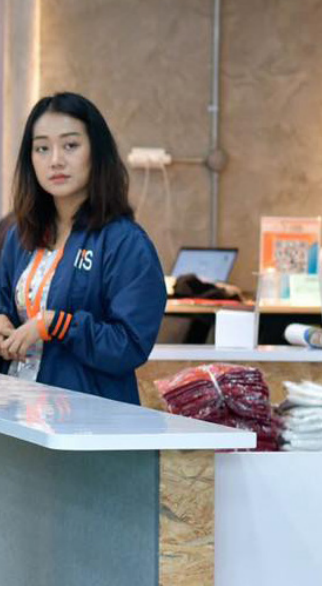
The Second Runner-Up: The New Past Story project from Sripatum University applies digital technology in narrating the stories of Thai and other cultures. The project aims to communicate with the new generation and encourage their learning experience in cultural and historical fields through online channels.

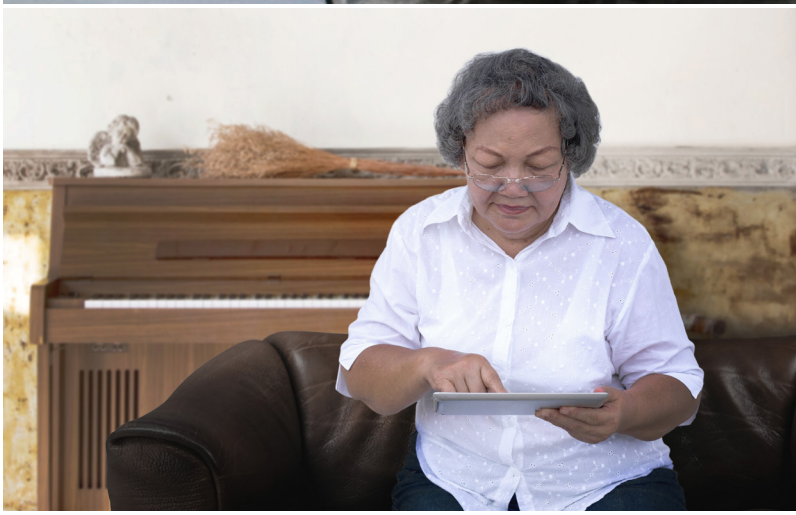
Thanks to these remarkable projects, the future of developing digital citizens of Thailand is brighter and more promising than ever. These citizens will not only be the main driving force of the country's digital transformation, but they will also steer the society and economy, creating more than 50 million baht in economic value.

Smart Cities that Fulfill Everyone's Needs

"Smart City Week Gen-Z Hackathon" is a collaborative event hosted by Thailand Smart City Office, Digital Economy Promotion Agency, and the NEXT– Creatorspace Center of KMITL Research and Innovation Service (KRIS). The event aims to nurture and develop innovators to become startup entrepreneurs in digital business, with 30 creative-minded Gen-Z individuals and urban development experts from Thailand and international agencies, namely Korea Trade-Investment Promotion Agency (KOTRA), Amazon Web Services (AWS), and City Possible Pioneered by MasterCard.

With the scope of innovation development limited to four areas in digital technology, biotechnology, logistics and energy, and creative economy, the challenge of this hackathon is to come up with an innovative design within 48 hours. Based on the "No One Is Left Behind" paradigm, the plan must address the people's needs, especially the visually impaired persons and residents of local slums. The developers are given information on the people's pain points with disabilities by a group of qualified speakers from the Thailand Association of the Blind (TAB), the Four Regions Slum Network, and the United Nations Human Settlement Programme (UN-Habitat). With such challenges, the Gen-Z developers must become Gen-Z hackers by coming up with the best solutions. Undoubtedly, they would need to have an in-depth understanding of the problem to create a cutting-edge prototype for the residents, based on the design thinking approach and empathy.





The prototypes developed by the Gen-Z talents help open our eyes to thought-provoking examples of Smart Living, which refers to the city development of Universal Design facilities to improve people's health and well-being so that they can live safely and happily in the future.

Smart City MEETs

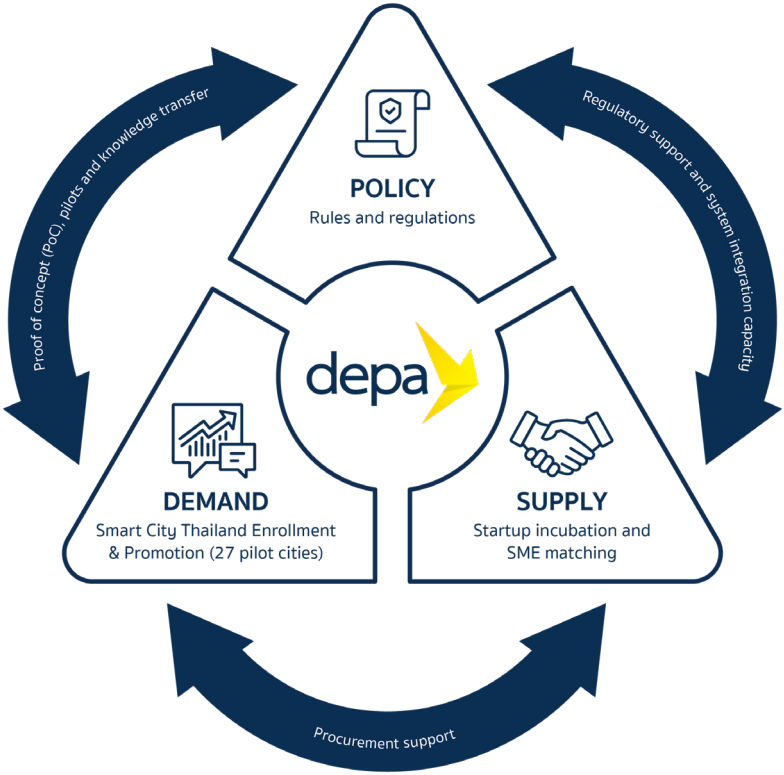
Digital Economy Promotion Agency as a Hub

Smart City Office Thailand was established in 2019 under the supervision of depa Thailand. It is a government agency that plays an essential role in promoting and supporting the development of smart cities. Moreover, it is the hub that connects people in smart city development, such as government agencies, the private sector, the education sector, foreign partners, and smart city areas in Thailand. Under the collaboration of these partners, the goals of Smart City Office Thailand are to create an agile ecosystem with strategic plans, apply technology and digital innovations to solve problems, and develop cities that meet the needs of everyone.

3 Supporting and Promoting Strategies of Smart City Development

As the bridge of smart city development projects, Smart City Office Thailand has come up with three strategies in three activities to build and support the smart city industrial market:

- The development of policies, regulations, and reward systems
- The creation of demand by building "Smart City Promotion Zone" around the areas supported by both the public and private sectors across the country
- The matching of services through the development of startup businesses and the support of SME businesses



Smart City MEETs

As the Center of Connectivity

Under the supervision of depa Thailand, Smart City Office Thailand remains central to various collaborations domestically and internationally. It is the stepping stone for those who want to be involved in the smart city development ecosystem or those looking for opportunities to participate, build or expand the smart city industrial market in Thailand. Thailand Smart City Week 2020 is an opportunity for future growth and plays a vital role as a hub of business connections in the smart city industrial market. As a result, the Smart City MEETS activity was initiated to exhibit digital technology and bring about solutions for smart city development from 50 entrepreneurs in urban development. It is the first virtual exhibition for Thai people to experience the technology of smart city development from every corner. Furthermore, it can drive the business matching opportunities through real-time conversations via https://scw.smartcitythailand.or.th/.

The information on digital technology and solutions from entrepreneurs that participated in the Smart City MEETs initiatives are as follows:

MEETs

The Governor of Khon Kaen: A Smart City Champion

What makes Khon Kaen special?

As one of Thailand's pilot smart cities, Khon Kaen aims at becoming the showcase of "a citizen-centric smart city" where technology meets local culture to improve the physical wellbeing and socioeconomic livelihood of the citizens.

What is Khon Kaen pursuing?

We value both local and international knowledge exchange, knowledge transfer, and proactive capacity building. We are particularly keen to embark on a Proof-of-Concept (or Poc) and the demonstration of the solutions that both the Handbook and this workshop may propose. The goal of this PoC is to test out the digital services for smart cities, especially in how they may be localized to fit the needs of the aforementioned citizen-centric Smart City of Khon Kaen. The benefits of the PoC also include the scalability of these services to other cities. We believe that Khon Kaen can be a great site for pilot projects. We believe in the benefits of digital technology which has proven to be effective in enhancing urban livability.

What's Khon Kaen known for?

Khon Kaen is internationally known for its "3M" smart initiatives – Medical (and Wellness), Mobility, and MICE (Meetings, Incentives, Conferences and Exhibitions) – such as healthcare such as the Khon Kaen Living Lab, Smart Ambulance, and Khon Kaen Mass Transit. Khon Kaen is working with many international partners such as the Ministry of Land, Infrastructure, and Transport of the Republic of Kore on the joint-study of the Mobility-as-a-Service system with the K-City Network International Collaboration Program and the British Embassy Bangkok through the Future Cities Program as well as the UK Tech Export Academy.

What are Khon Kaen's Long-Term Goals?

First, to enhancing resiliency and sustainability for Khon Kaen's industry by creating an integrated data-oriented platform in all domains of a smart city; and, Second, to scaling up the successful "Khon Kaen Smart City '' model to other cities, utilizing effective and cost-saving digital big data-oriented platforms to further enhance Thailand and ASEAN's smart city ecosystem.

Dr.Somsak Changtrakul
Governor of Khon Kaen Province

Frequently Asked Questions (FAQ)

Q: Does it only have to be a province to be developed into a smart city?

A: No, two types of areas can be developed into smart cities: (a) the development area at the provincial/ district/sub-district or municipality levels and (b) the area developed by the private sector or other government agencies.

Q: Is it necessary to develop a solution that covers all seven areas of smart city development?

A: No, it is not necessary to provide services in all seven areas, but at least two areas are required, i.e., Smart Environment and any other area depending on the context of the space.

Q: What would be the benefits of becoming a smart city?

A: The private sector that invests in the development process will receive an 8-year tax exemption privilege (with limitations) from the BOI. Details can be found at

Q: How long does a Smart City badge last?

A: It is valid for two years. However, the applicants must report the progress every six months within two years.

Q: What are the criteria for becoming a smart city?

A: There are five criteria:
1. Having clear goals and precise areas of development
2. Having guidelines for urban infrastructure development
3. Having a secure city data collection and management system
4. Having smart city services in two of the seven dimensions of smart city
5. Having a sustainable management approach

“Smart City”

is a word that everyone might be familiar with. However, not everyone truly understands its underlying meanings. What is a smart city? What does a smart city look like? Can smart cities genuinely improve the quality of life for urban residents?

Climate change, pandemics, the growth of populations, and mega-urban migration have accelerated the pace of development in many megacities. Thus, preparing and handling these situations have become an essential and urgent matter in different areas, such as infrastructure management, environmental and housing management, social services, public health services, and education. Given such challenges, traditional technology might not be adequate to carry out innovative solutions. At the same time, to bend with the force of urbanization, digitalization has become a new dimension for urban citizens. Therefore, smart city development is one of the resolutions that can help many cities cope with their concerns and serve as another crucial factor of smart city development.

Definition of “Smart City”

Smart City is a city that capitalizes on modern technology and digital innovation to enhance the efficiency of city services and management systems. Such technology will help reduce costs and maximize resources, emphasizing a friendly design and active engagement between the business and public sectors. All in all, the main goal of developing a smart city lies in the creation of a livable and modern city, which, in turn, will improve the quality of life of the people and generate sustainable happiness for all.

Livable City

Through technology and innovations, a livable city refers to a city developed in a variety of fields to fulfill the diverse needs of the residents. For instance, basic infrastructures, public utilities, housing, resting areas, worksites, commercial spaces, and the arrangement of the city’s space should be developed in the most compatible way with the city’s identity, culture, and traditions. Therefore, this type of smart city development focuses mainly on addressing pain points according to the city’s needs.

New City

New City refers to a city that is completely redeveloped and transformed into a modern city. Similar toLike the livable city concept, technology and innovation remainremains at the core of the development, covering seven areas of the smart city development agenda. In other words, the city should be developed in all fields, including basic infrastructures, public utilities, housing, resting areas, worksites, commercial spaces, and the proper allocation of city spaces. An example of this type of smart city is Wang Chan Valley in Rayong Province, which was built from scratch by PTT. Specifically, the city was completely transformed from a no man’s land into a brand-new capital.

7 Smarts

7 Dimension of Thailand Smart City

A city that takes into consideration the environmental impacts and climate change conditions. It focuses on using technology to systematically support water management, climate, waste management, and disaster surveillance and increase people's engagement in conserving natural resources.

A city that can manage energy efficiently, balancing between the production and energy usage to foster energy security and reduce interdependence on the power from the main grid.

A city that develops government services to better facilitate the people and stakeholders in accessing government information and services. It places a great emphasis on transparency, participation, and consistent improvement through service innovations.

A city built by taking into account the Universal Design, thereby improving the people's health, safety, and well-being.

A city that uses digital technology to create added value in the economy and manage resources efficiently. Examples include smart agricultural cities and smart tourism cities.

A city that focuses on developing smart traffic and transportation systems to drive the country. Moreover, it increases the efficiency and connectivity of various transportation and traffic systems, improves the convenience and safety of transportation, and cares for environmental impacts.

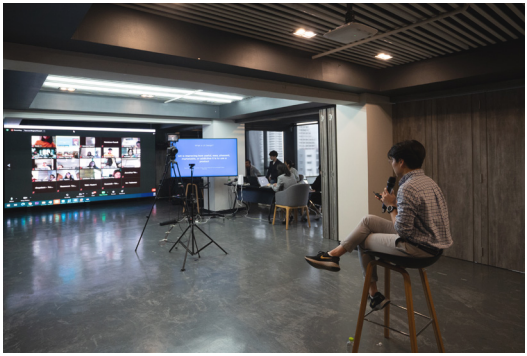
A city that aims to cultivate knowledge, improve skills and build an environment conducive to lifelong learning. Moreover, it intends to reduce social and economic inequality and foster creativity, innovation, and participation.

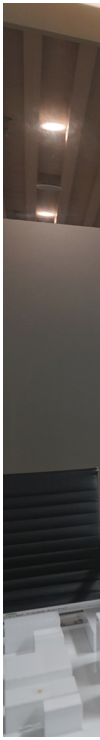
Special Article:
The Making of an Innovative Governmental Organization and the Development of Smart Cities in Thailand

The Importance of Cities and Innovation Development

According to the United Nations, a "city" is recently redefined as an area where more than 55 percent of the world's inhabitants reside. The cause of this change in definition is a result of the change in economic systems, particularly the replacement of traditional agriculture and manufacturing sectors with the services sector that generates higher economic value. Thanks to this, the economy has grown by leaps and bounds. Furthermore, coupled with technological advancements, machines and automation can now, in some part, replace the work of humans with higher precision and better efficiency than human labor in agriculture. As a result, the traditional manufacturing sector is dwindling. Both contexts are the push and pull factors that cause the movement of urban migration for severala number of reasons, particularly working, studying, or residing. As a result, this forms the centralization of infrastructure, economic, social, and cultural investments of the urban areas—all of which can be denoted by the density, compactness, and cultural diversity.

This "urbanity" is what many thinkers and researchers regard as the source of creativity, an essential element of innovation, particularly of technological change. Provided that a city is an area filled with expertise, common problems, density, and space constraints, ideas and interests are being gathered and transformed into solutions, which are the main driving force of the economy or the so-called innovative ecosystem. Undoubtedly, the continuous development of innovation fulfills the needs of the people

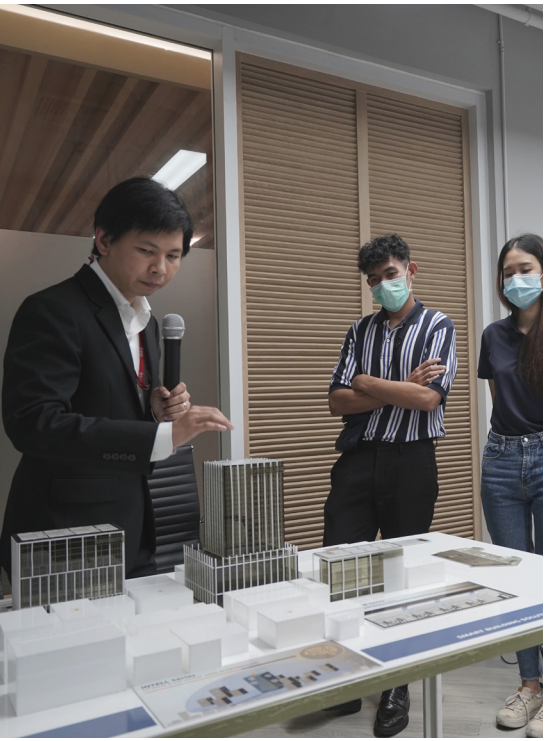




during different times, eras, and contexts of society. Therefore, almost all innovations that have influenced our lives from the past to the present stem from this "urbanity."

A city with a better environment and higher equality would positively influence innovation development. Provided that the number of urban inhabitants in the world will rise about 68 percent (more than two-thirds of the world) in the next three decades, "the future of cities, therefore, is the future of innovation development." The development of such innovations is seemingly the future of the majority of people. In Thailand, if we analyze the Thai economic system, we could witness that the importance of the cities has increased over time in almost every area, such as society, ways of living, economic development and commerce, and geopolitics of Thailand, to name a few.

Apart from being a place of incubation, a city can also be a source of pollution and negative externalities. Often, the development of a city worsens environmental issues, mainly due to



the reckless use of resources by urban residents and economic activities. For example, urban buildings usually consume more than half of the total urban energy. As long as the business system is service-related, these buildings would consume a considerable amount of power for a long time.

In addition, the source of electricity stems from the production of unclean energy with poor circulation. Thus, this wasteful use of electricity directly accelerates the greenhouse effect and generates various pollution problems such as dust, air, and water pollution. Moreover, it also leads the urban residents to turn into global consumers, causing harmful environmental impacts that may take centuries to rebuild a livable commonplace for all.

Innovation and Smart Cities

The development of city potential is both supporting and nurturing factors of innovation incubator and ecosystem—both of which are important to drive innovation and are the main components of city development. Smart City is one of the approaches that use technology in city management to fulfill the needs of the residents. Different types of technologies are needed to build a Smart City, but most importantly, big data. Specifically, it can be used to analyze and translate raw data into a set of information that facilitates decision-making in the most precise and targeted manner.

One crucial point to note is that a smart city is not a fixed output but rather a process that brings out the potential of a city through the continuous development process. It starts with setting up a framework for the development process to be carried out in stages later. However, for a city to be fully developed into a sustainable smart city, it should consist of these components:

1) The boundaries and vision of the city are based on the economic, social, environmental, cultural, and other relevant factors of the area;

2) Smart city infrastructure investment approaches cover basic digital infrastructure, transportation infrastructure, energy infrastructure, public utilities, and any other infrastructure corresponding to the goals, objectives, forms, and types of the smart cities;

3) Guidelines on the development of City Data Platform (CDP) by linking or using information for management. This also includes providing services in smart city areas, cybersecurity management guidelines related to smart city systems, and the safety of personal data;

4) Details of the of smart city services, including activities or projects corresponding to the type and charac-

teristic of smart city development as requested for consideration under (1); and,

5) Sustainable management guidelines of smart city development in the form of public sector only, or a hybrid of private and public sectors, or other private sectors as designated by the Smart City Development Steering Committee.

Given that each city has different needs, readiness, and willingness to develop may be the first factors to consider when deciding to become a smart city. Overall, Thailand's Smart City policy is aimed at guiding Thailand toward becoming a developed, high-income country where people can enjoy a high quality of life with less inequality in all aspects. Based on such expectations, governments recognize the importance of digital technology by prioritizing people's digital literacy and assisting them in effectively accessing digital technology.

It is important to note that "Smart City" development is only a process, not a technological outcome. Instead, the most important technology provides measurable livelihood to citizens. Therefore, it is essential that a city develops its framework and prioritizes the needs of its citizens. Furthermore, if a city can accommodate the needs of the residents, there will be less investment cost, thanks to the efficient allocation of resources; otherwise, this development process could be overpriced. Thailand can learn from both lessons to determine which studies would maximize benefits as we advance.

Promotion of "Smart City" by the Governmental Sector

Currently, Thailand is striving for "Thailand 4.0 Policy," which aims to drive a value-based economy rather than a quantifiable one. Therefore, innovations that can effectively fulfill economic, social, and environmental should follow a "Bottom-Up" development approach or a people-centric framework. With this model, if the cities grow, the country would rise accordingly. Today, the development of a smart city is led by the National Steering Committee on Smart City, appointed by the Prime Minister of Thailand and chaired by the Deputy Prime Minister. They have begun to implement the smart city development plan and tasked depa Thailand as the Committee Secretariat to set the criteria for the smart city development.

Applications can be submitted to depa Thailand to receive special promotions and benefits offered by the government, such as tax benefits offered by the Thailand's Board of Investment (BOI), Sandbox benefits (Sandbox refers to an experimental area that can freely apply technologies) provided by the Office of the Broadcasting Commission and the National Telecommunications Commission (NBTC). However, the word "city" is not always restricted to government agencies. Any private sector that can offer spaces in industrial estates, educational institutions, or any cooperation between the private sector and the city can submit a proposal to depa Thailand to receive support for developing a smart city. Typically, the goal of Thailand is to reach 100 cities by 2022--47 of which already have been implemented a smart city development plan. More information can be found at www.smartcitythailand.or.th

Despite clear visions, active government support, and innovation development, the development of "Smart City" in Thailand has opportunities and challenges. As such, the leading role of depa Thailand is to ensure that those challenges are addressed, and opportunities are protected. Furthermore, provided that the

development of smart cities is relatively new, depa Thailand's primary responsibility is to create shared understanding across all sectors, including both on the demand side (or the city itself), and on the supply side (or the producer of technology). Thus, such a challenge will allow both the supply and demand sides to collaborate, experiment, and test out the technology.

Furthermore, Depa has designed a course named "Chief Smart City Officer" (CSCO) to build smart city developers. The course focuses on building practical knowledge such as key concepts, rationale, and digital technology in urban development. Moreover, it will assess the project's feasibility as the development frameworks usually involve new investments, such as establishing a city development company and a Public-Private Partnership (PPP). For those who are interested, CSCO is now open for registration. The course would begin in August 2020. To inquire for further information, please contact depa Thailand's Smart City Promotion Department at scp@depa.or.th.

Finally, to raise awareness of smart, livable cities both in Thailand and abroad, depa Thailand organized "Thailand Smart City Week 2020, which took place during October 2020. This event is part of the strategies that promote knowledge sharing, particularly among cities aiming to become smart cities and technological providers from Thailand and abroad.

Furthermore, the event intends to incorporate digital innovation into smart city development in order to improve people's quality of life. Furthermore, Thailand Smart City Week serves as a platform for Thailand to become the global leader in driving smart cities, particularly in the Southeast Asian region. Last but not least, the event is committed to encouraging collaboration in the development of smart city projects, particularly through public-private partnerships, and to continuing to build a stable learning platform, such as the CSCO course in smart cities.

Dr.Supakorn Siddhichai
Group Executive Vice President
Digital Economy Promotion Agency

Smart City Promotion Zone
47 Cities

7 Phitsanulok Smart City
8 Phitsanulok Safe and Smart Tourism City
9 Naresuan University Smart City
10 Udon Thani Smart City
11 Khon Kaen Smart City
12 Mukdahan Smart City
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35 Phuket Smart City
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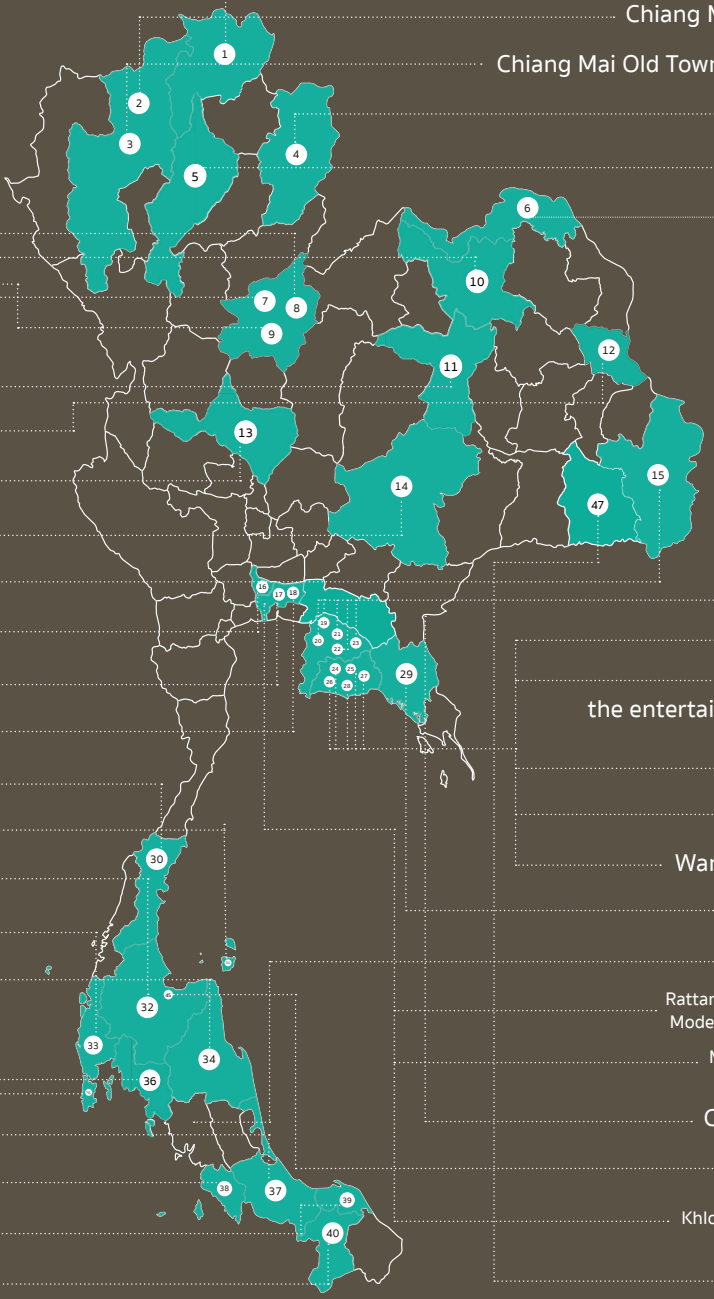
37 Hat Yai Green Smart City 2567
38 Satun Smart City
39 Pattani Smart City
40 Yala Smart City



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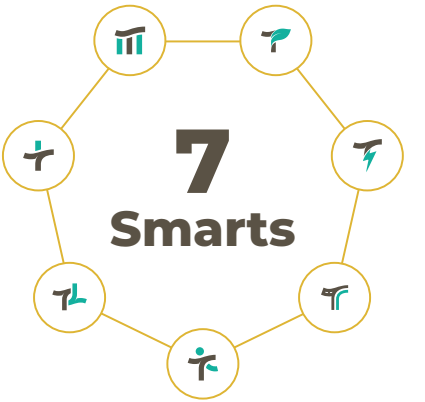
Livable City
New City

5 Pilot Smart Cities
Smart City Thailand

In light of the assessment of the seven dimensions of smart city, the working group has put forward a smart city development plan to the National Smart City. Sub-Committee and National Smart City Steering Committee. The first batch of the qualified cities are:

Phuket
Smart City

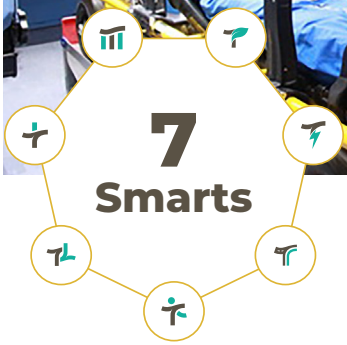
The goal of Phuket Smart City is to be a sustainable city for tourism, education, innovation, and services at international standards.



Phuket will have smart city services and solutions in all 7 dimensions, with a total of 42 smart city projects. Among these, flagship projects include the comprehensive integration of the artificial intelligence-powered closed-circuit cameras (CCTV), serving as the city's robust security surveillance system. To help reduce crime and prevent accidents, both face and license plate recognition systems will be concurrently deployed. Promoting community-based tourism through the Phuket Tourism Information Management system will also enable service providers to boost demand and attract visitors. In addition, there are also digital projects for the tourism of the adjacent Andaman coast, i.e., the so-called "Andaman Tourism Digital Twin" project which will help to distribute growth in the sustainable tourism economy throughout the region. Renowned legacy projects such as the Phuket Smart Bus will continue to operate synergistically with novel projects like the Mobility-as-a-Service (MaaS) platforms to serve both the residents and visitors.

Khon Kaen
Smart City

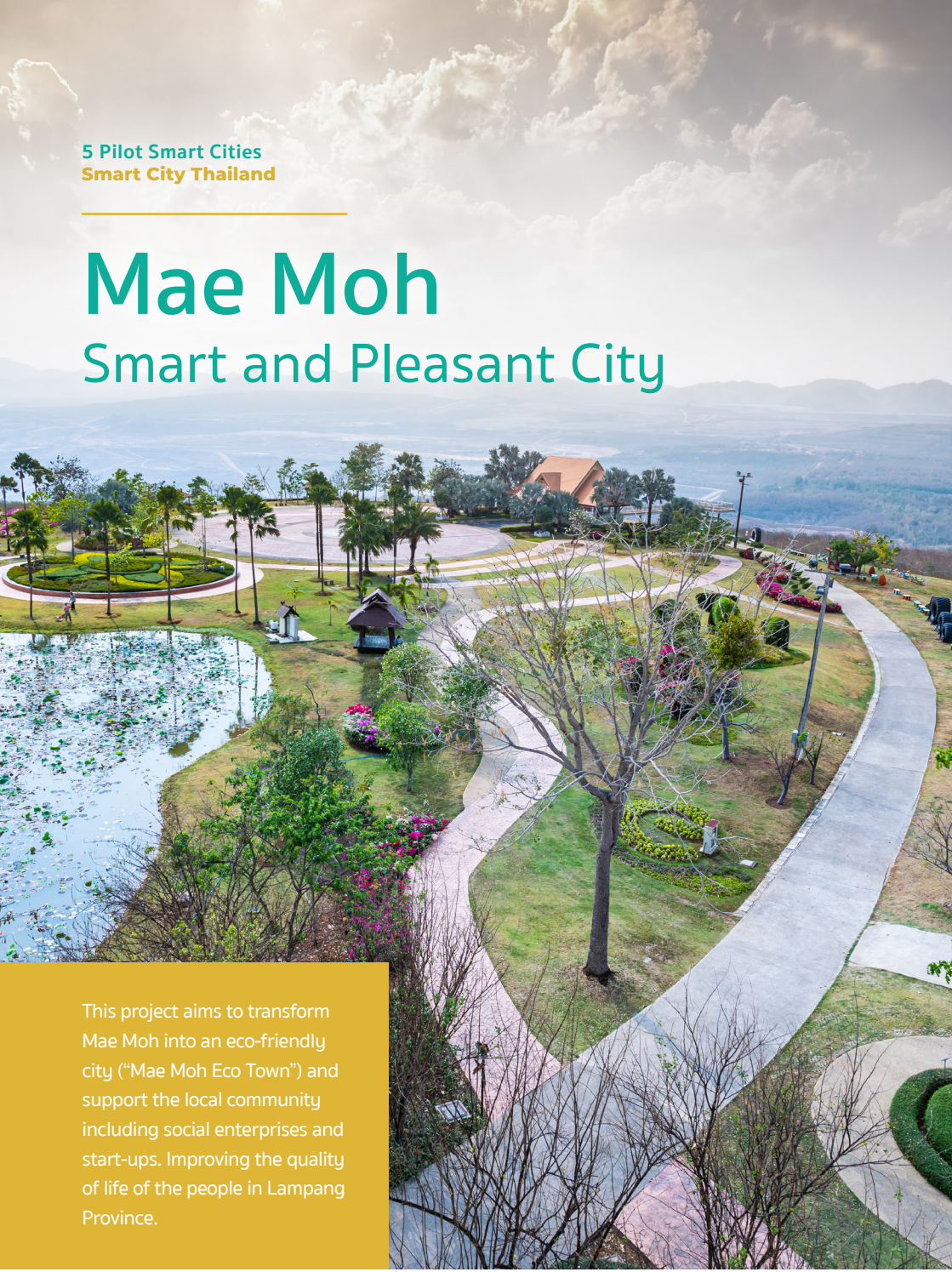
This project aims to develop itself to become a livable metropolis and an innovative city that connects all economies in the Greater Mekong Subregion (GMS).



The total number of smart city services that will be in operation stands at 28, covering all 7 dimensions. Notable projects include the maximization of green areas by using the Internet-of-Thing (IoT) technology to manage ecological resources and mitigate community waste problems. By using IoT sensors to monitor the amount and condition of waste as well as provide complete analytics of the collection route, the streamlining of this end-to-end waste management process can enhance the already-world-renowned urban landscape and improve the wellbeing of the citizens. In the domain of the smart economy, Khon Kaen will continue to build an ecosystem for entrepreneurs, especially SMEs and Startups in the IoT or digital industry. Medical and healthcare applications for comprehensive public health services are being designed to foster the city's 3M value proposition: Mobility, Medical and MICE (Meetings, Incentives, Conferences and Exhibitions). The connectivity of transport networks through information technology, such as the light rail transit, is staged to become an important component of the city's growth through transit-oriented development (TOD).

Mae Moh

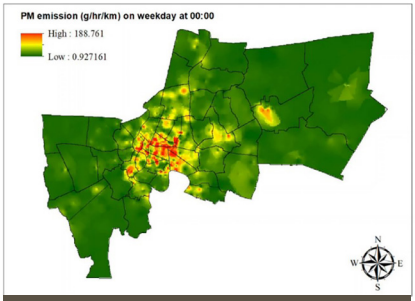
Smart and Pleasant City



This project aims to transform Mae Moh into an eco-friendly city (“Mae Moh Eco Town”) and support the local community including social enterprises and start-ups. Improving the quality of life of the people in Lampang Province.

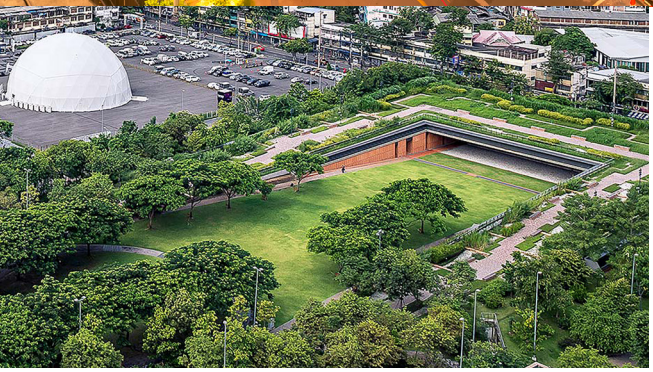
3 Smarts

Mae Moh will have smart city services and solutions in 3 domains, with a total of 10 smart city projects, including those of smart environment and smart energy which are EGAT's general expertise. The flagship projects of "Mae Moh Smart and Pleasant City" include rehabilitating 20,000 rai (7,907 acres) of community forests, building 100 water management prototypes, and introducing robust climate sensors. State-of-the-art Geographic Information System (GIS) will prompt results for effective environmental management. Moreover, digital payment services will be introduced to generate income for the community by promoting and community products. As an Eco Town, it will adopt a smart grid technology using a smart metering system and installing solar power generation on the roof to reduce the use of fossil-based electricity and therefore increase the use of renewable energy.



Smart Environment

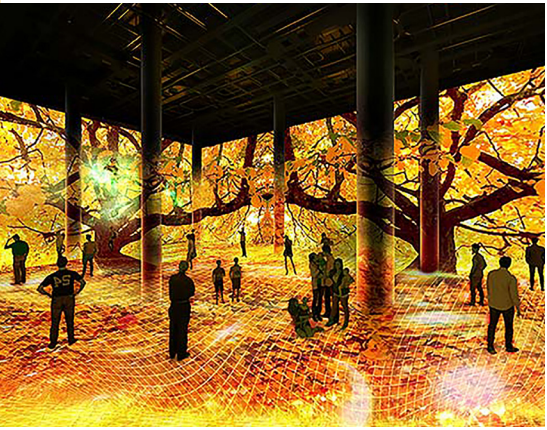
A city that takes into consideration the environmental impacts and climate change conditions. It focuses on using technology to systematically support water management, climate, waste management, and disaster surveillance and increase people's engagement in conserving natural resources.

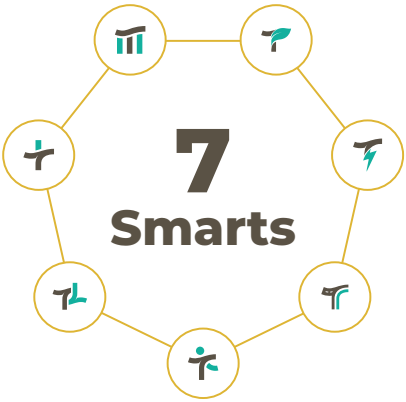


5 Pilot Smart Cities
Smart City Thailand

Samyan
Smart City

Samyan Smart City is operated by Chulalongkorn University, Thailand's most prestigious university. It is natural for Samyan to aspire to become an "Innovation District," creating both innovation and shared values for communities and Bangkok's urban society.

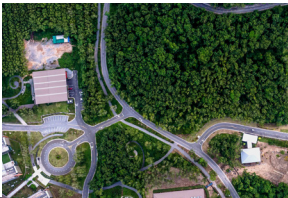




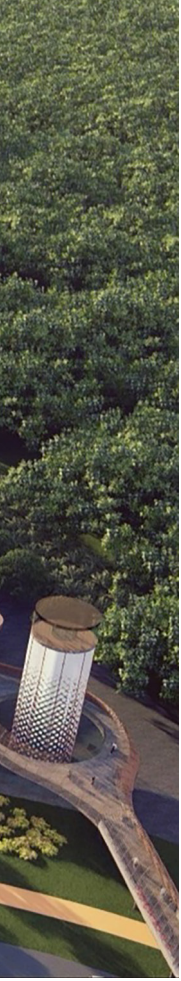
There are going to be 42 smart city projects across all 7 domains, including a comprehensive water management system and wastewater recycling system (both systems are set to be built around the award-winning Centennial Park). Furthermore, the Centralized cold room control system and area energy management system (AEMS) will enhance the efficiency of the overall management of both energy production and consumption. The use of renewable energy through solar roofs and solar parking will become a norm of this eco-urban campus and vibrant community. In addition, the "Creative & Startup Village " is a conceptual pilot project aimed at creating Thailand's first innovation district, to be located at the historical Block28 where the legacy of the Old Samyan Market stands. There will also be a variety of end-to-end, door-to-door, mobility-on-demand services, ranging from electric vehicle (EV) car-sharing, bike-sharing, Tuk-Tuk-sharing, and E-Scooter-sharing, as well as an EV car rental platform. Such services will effectively increase travel options and enhance the efficiency of public transport, while also reducing road accidents and greenhouse gas emissions within the area.

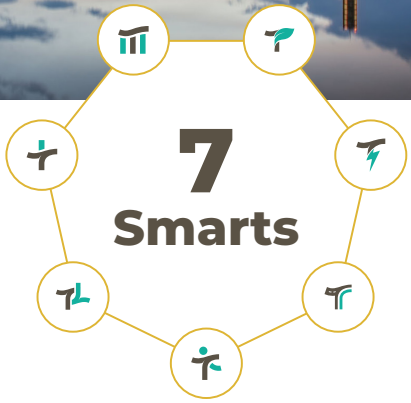
5 Pilot Smart Cities **Smart City Thailand**

Wangchan Valley
Smart City

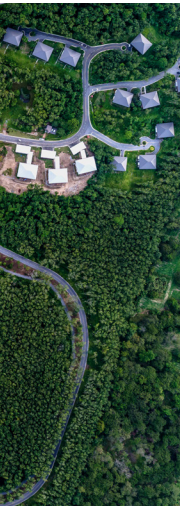


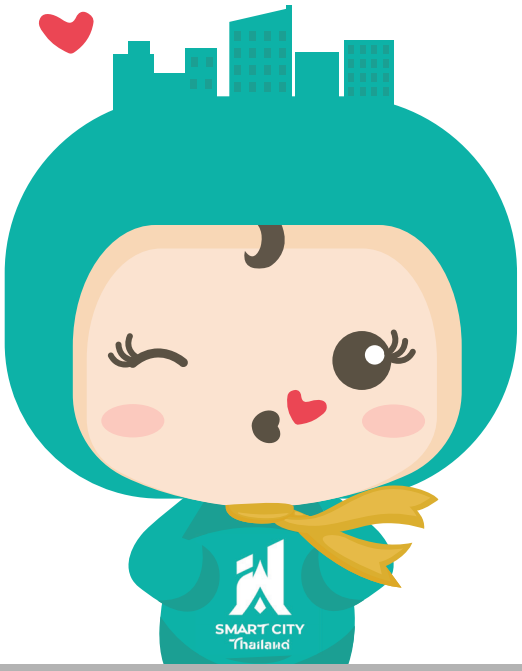
Wangchan Valley Smart City will be built around the value created by research and development for innovation (R&D). This new city will have infrastructure that meets international standards, allowing people to live in a better quality of life and in a healthier environment. Located in Rayong Province's scenic Wang Chan District.





There will be smart city systems across all 7 domains, totaling a remarkable number of 55 projects. The flagship projects include a comprehensive solid waste management system, real-time weather monitoring system, air quality monitoring system, and energy management system (EMS). As a national energy provider, PTT shall utilize this site to promote its renewable energy production using big data analytics. Additionally, innovation is welcomed through the research and development sandbox, world-class innovation center, intellectual property consulting center, and an innovation incubator for SMEs and startups. There will be an electric bus service (EV Bus and EV Shuttle Bus) with an integrated CCTV camera system that provides real-time monitoring of driving behavior and connects to an automated dispatch system for the emergency ambulance and remote medical services (also known as telemedicine).





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